



A LEGO® BRICK MODEL OF

VHS Cassette

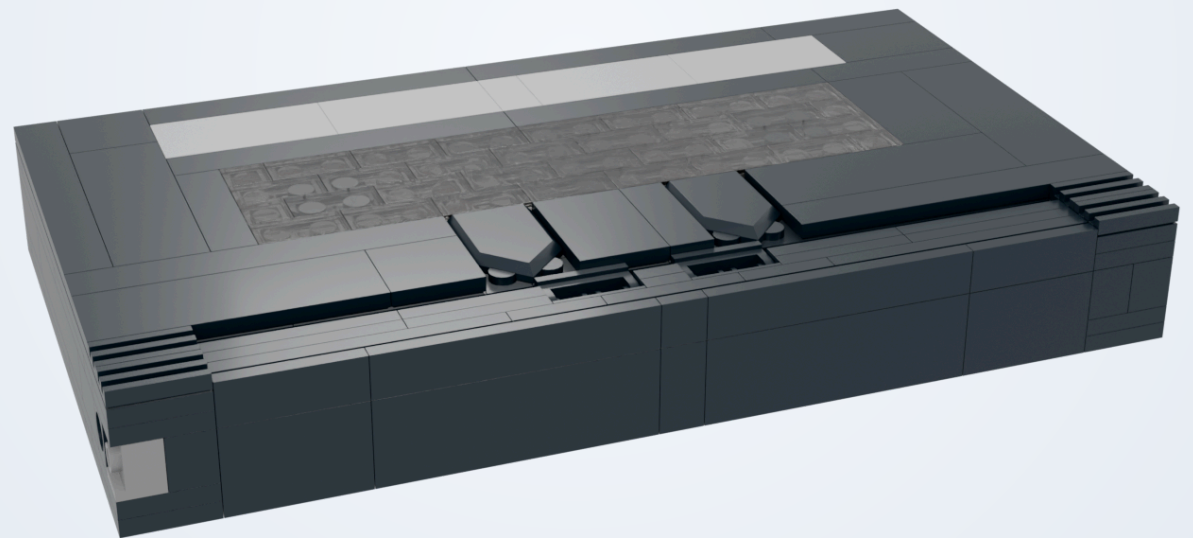
A 1:1 VHS tape, within a few millimetres of the real 187 x 103 x 25 mm cassette. Flip the dust door open; spin the reels through the window or from underneath.

348 pieces · 81 steps · 6 sections

Unofficial fan-made building instructions

Not physically build-tested. Every part is a real LEGO element and the model was checked on a computer (connections, collisions, every step, balance, mechanism), but nobody has built it with real bricks yet.

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Before you start

An unofficial fan-made manual

This booklet is not a LEGO product. It was generated from the model file by a computer program and checked part by part.

Not physically build-tested

The model and these steps were checked on a computer only: every part/colour is a real LEGO element, all 899 connections between parts were matched in 3D, no two parts overlap, every part can be pushed into place in its step, and the model's balance was estimated; the mechanism was swept through its whole range. Not tested: how firmly parts hold, part tolerances and how it copes with handling. Take your time and support the model while you work.

The finished model

About 18.4 × 10.4 cm and 2.6 cm tall, about 288 g (estimated).

Two black halves close around two reels: a full supply reel and a nearly empty take-up reel, each spinning on its own pin behind a smoked window. The dust door clicks onto two hinges and swings up like the real flap. Look underneath for the drive holes, the screws, the write-protect tab and the open mouth where the head drum goes in.

You will need

- The 348 pieces in 7 colours listed at the back, with their LEGO element IDs and BrickLink numbers.
- A brick separator for pieces that sit too deep for a finger.

Tips

- Sort the pieces by colour before you start.
- Build the sections in order: each sub-assembly is finished before it is used.
- New pieces in a step have a yellow outline; earlier pieces are drawn paler.
- The dust door hangs on two click hinges: press it on firmly until both hinges click.
- The reels sit loose on their pins until the top half closes over them. Keep the halves level while you lower the top.

The 7 colours

 Black: 206	 Flat Silver: 11
 Trans-Brown: 79	 Light Bluish Gray: 3
 White: 31	 Dark Bluish Gray: 1
 Reddish Brown: 17	

How to read the steps

Parts and step number

7



2x

The parts for the step and how many of each, then the step number. All parts are drawn at one scale.

New parts

The parts added in a step have a yellow outline and are drawn in full colour; the parts from earlier steps are drawn paler.

Sub-assemblies



1x

A yellow-framed picture means: put in the sub-assembly you built in its own section. "Build 4x" means you need four of them.

Sections

Each section starts with its name at the top of the page. Sections are built in the order they appear.

Seen from below

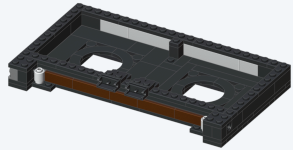
When a step adds parts underneath, the picture shows the model from below. A small label says so.

Insert direction

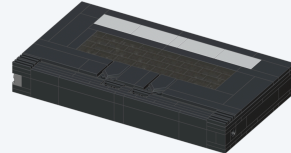
Some parts slide in sideways (axles through bearings, gears onto axles). The picture shows where they end up; push them in along their axle.

Build overview

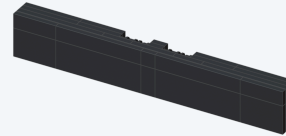
The model is built in 6 sections. Each is finished before it is used.



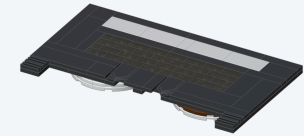
1. Bottom half
steps 1–35



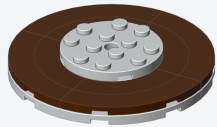
2. VHS Cassette
steps 36, 45, 81



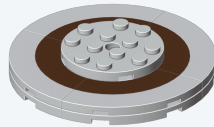
3. Dust door
steps 37–44



4. Top half
steps 46–70, 75, 80



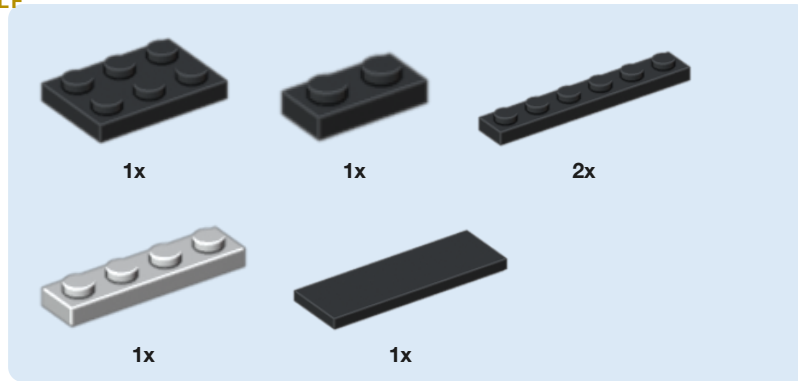
5. Supply reel (full)
steps 71–74



6. Take-up reel (nearly empty)
steps 76–79

BOTTOM HALF

1

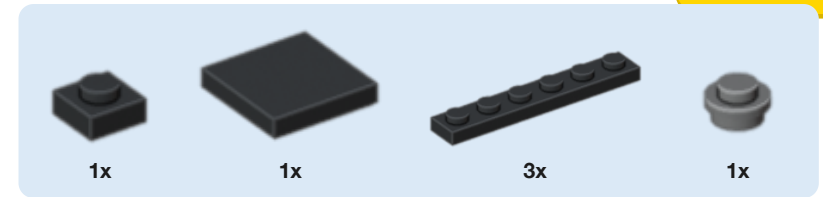


Bottom half: floor plates, tied together by the tiles and the first wall course

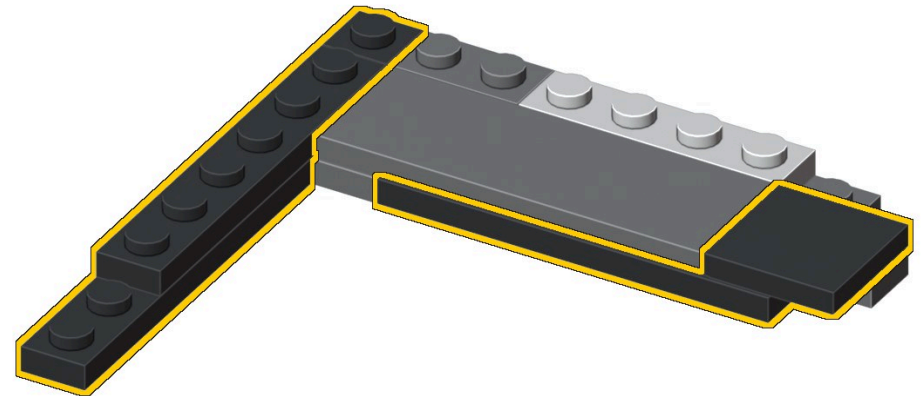


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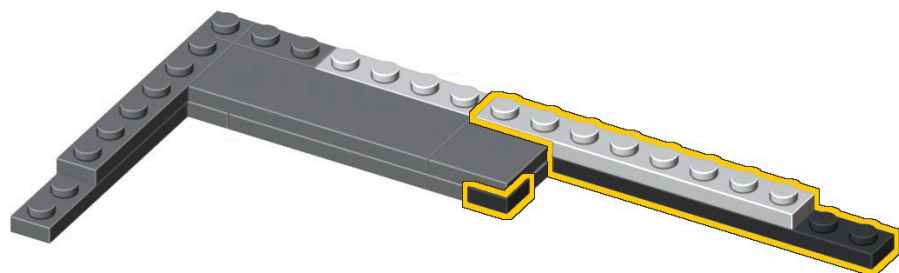
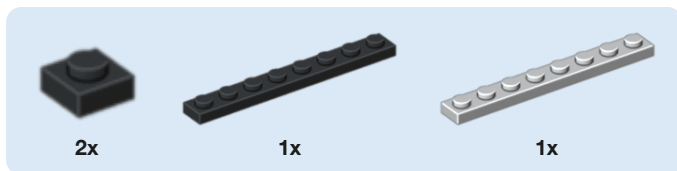
Bottom half



Silver screw heads sit in the floor: five of them, like the real thing

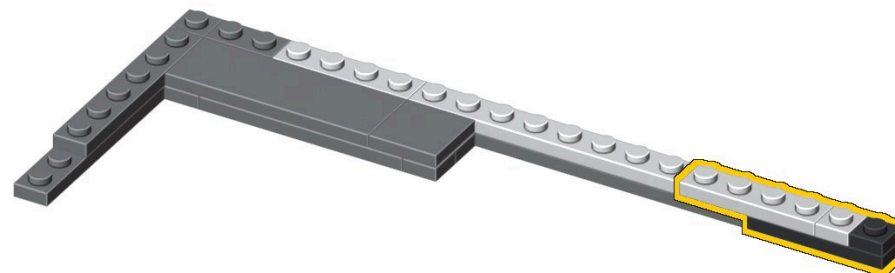
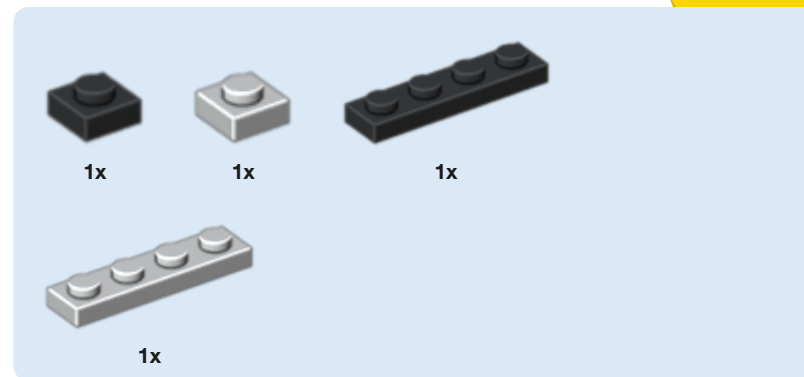


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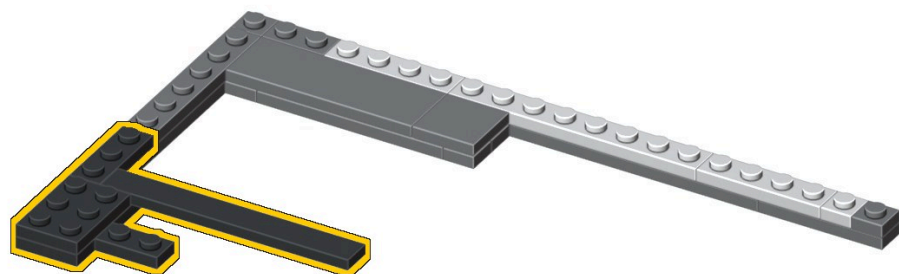
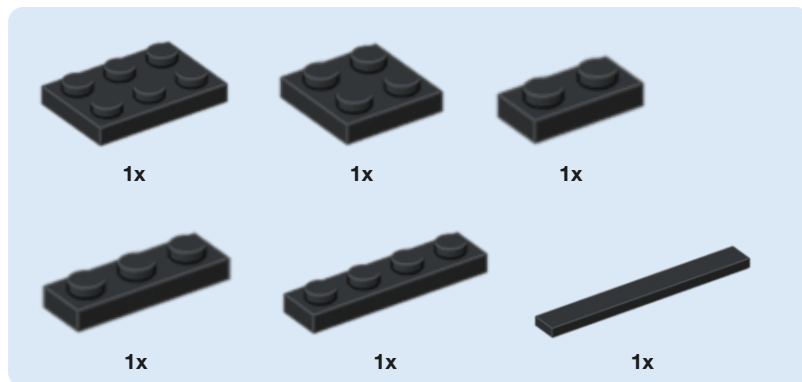


4

Bottom half

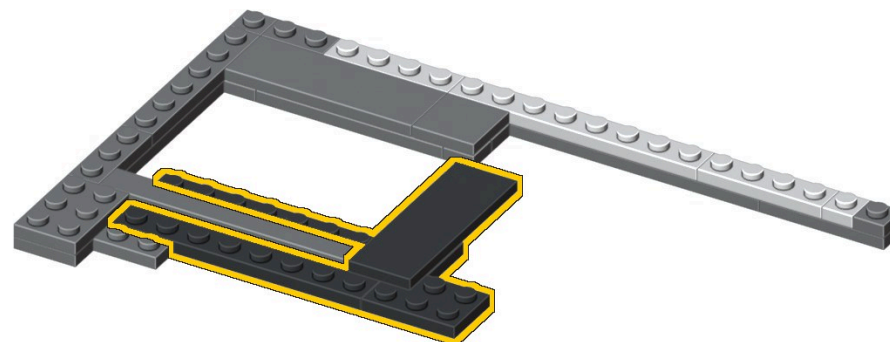
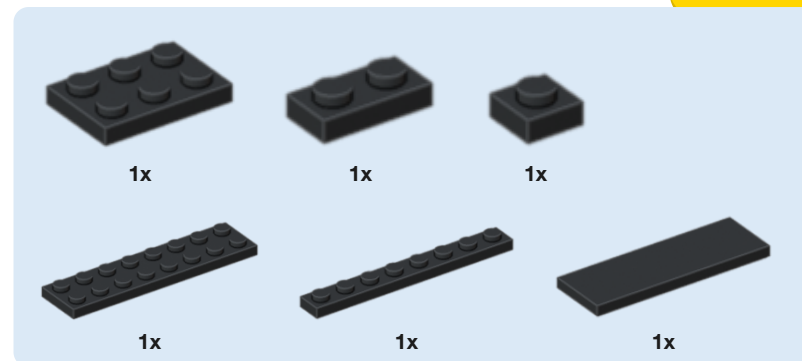


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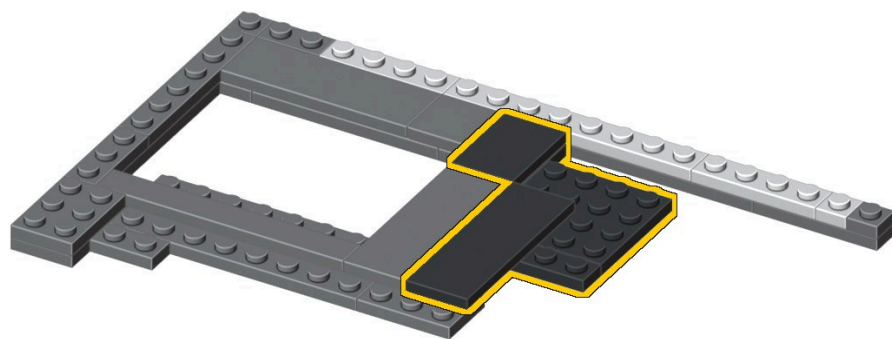
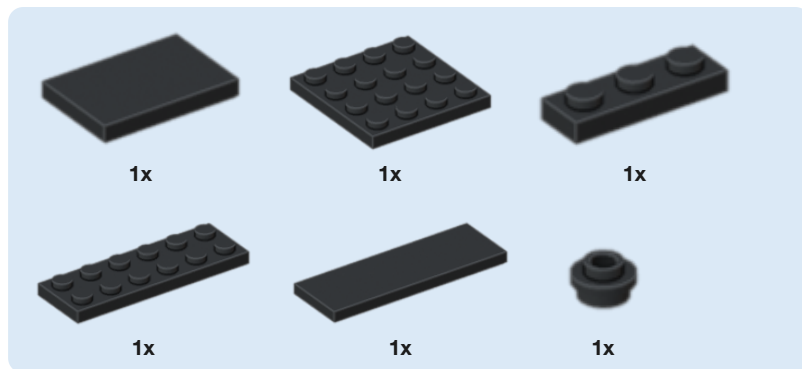


6

Bottom half

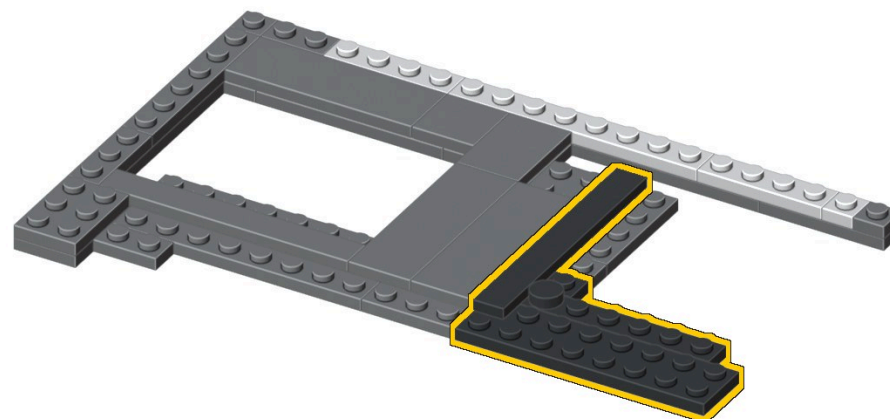
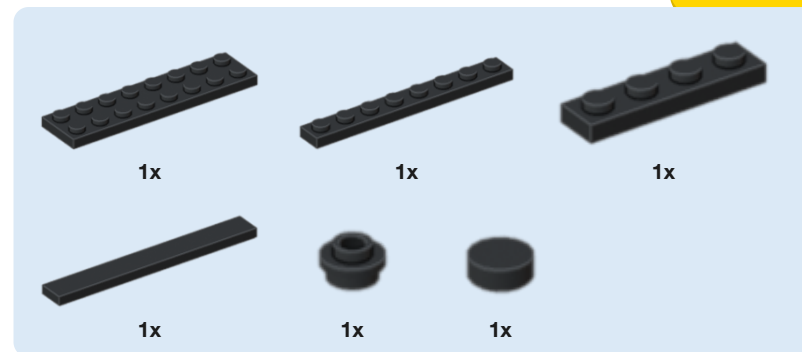


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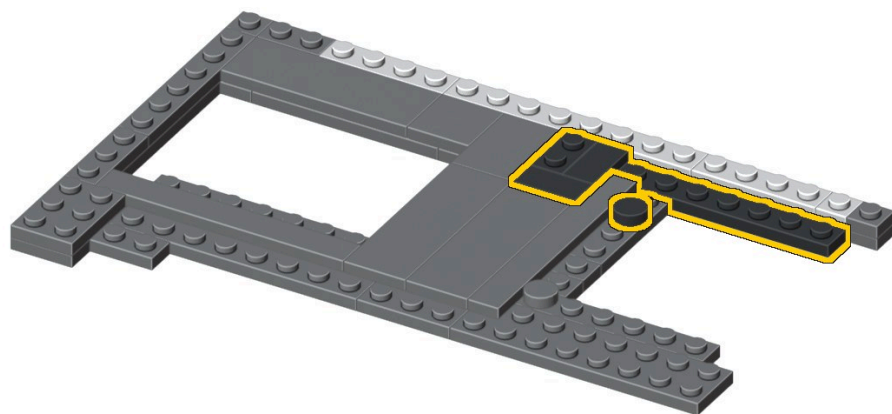
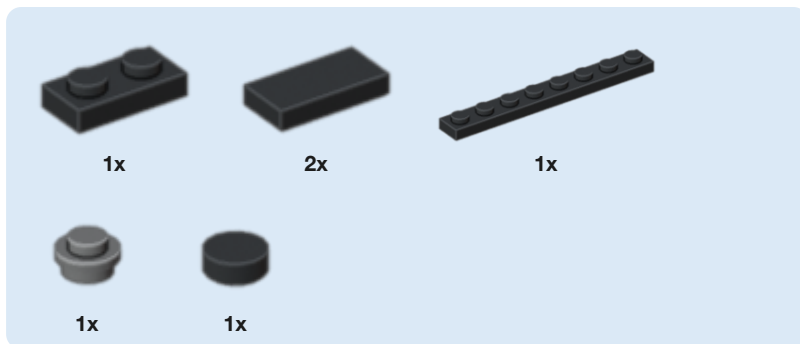


8

Bottom half

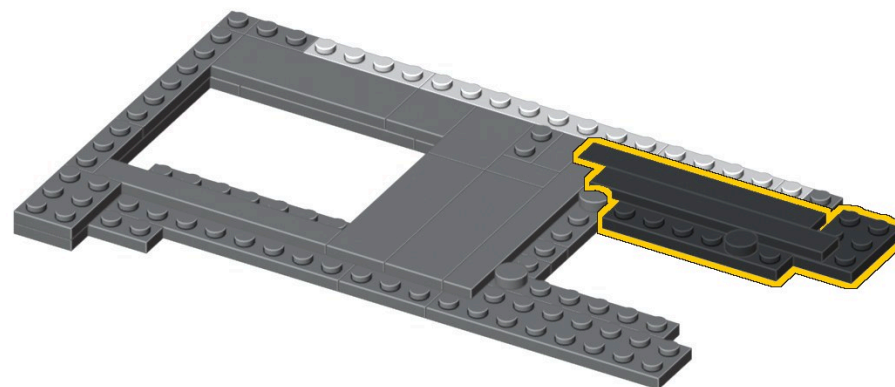
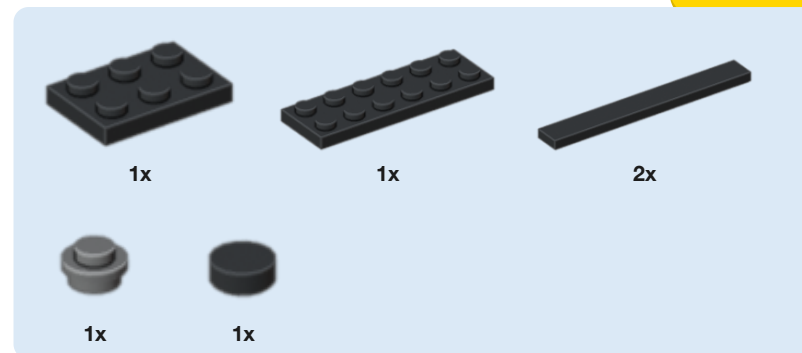


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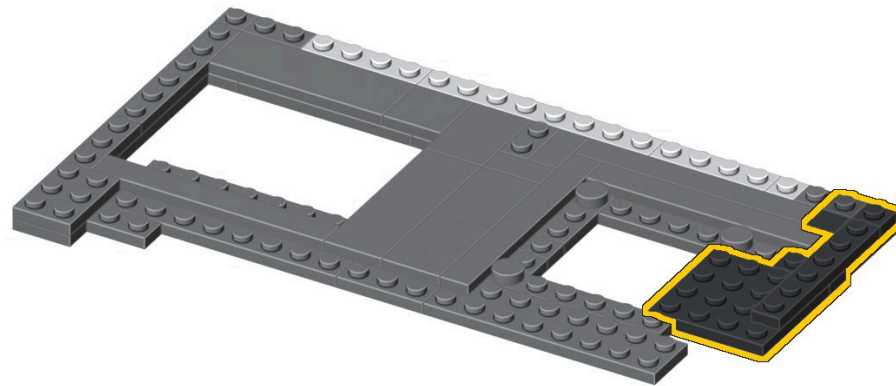
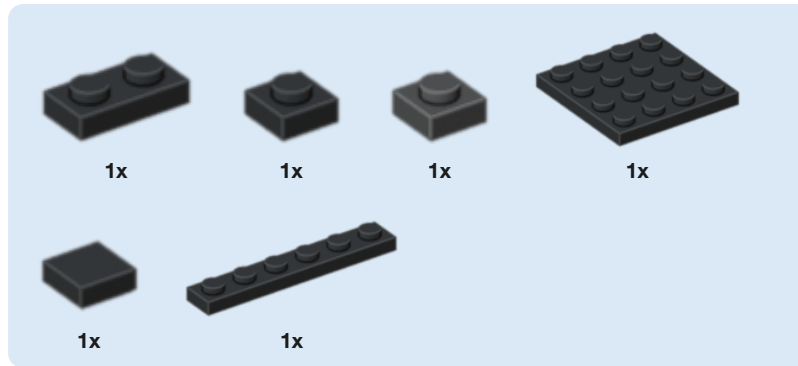


10

Bottom half

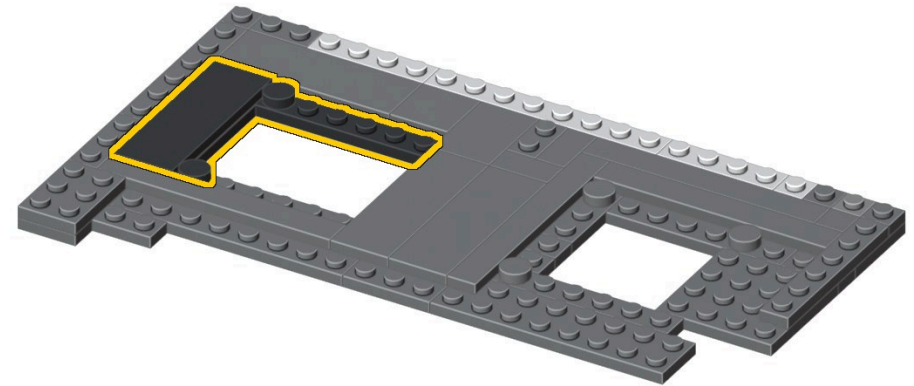
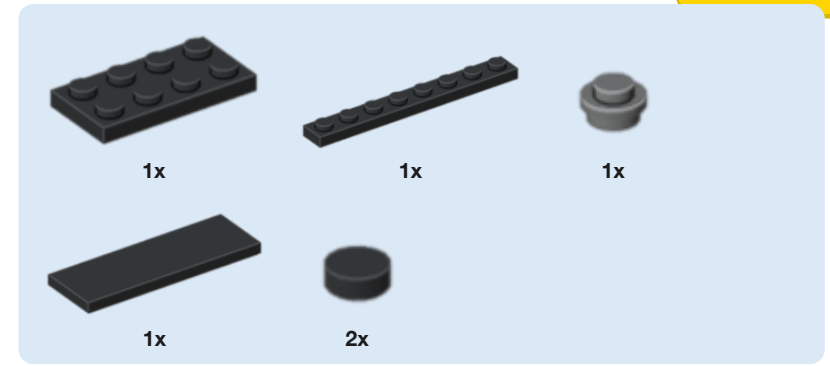


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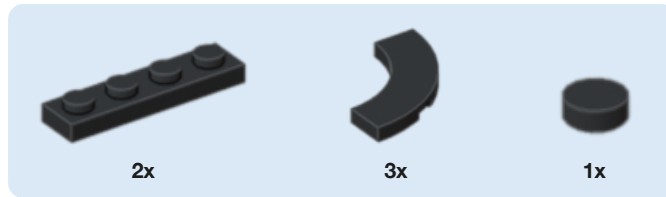


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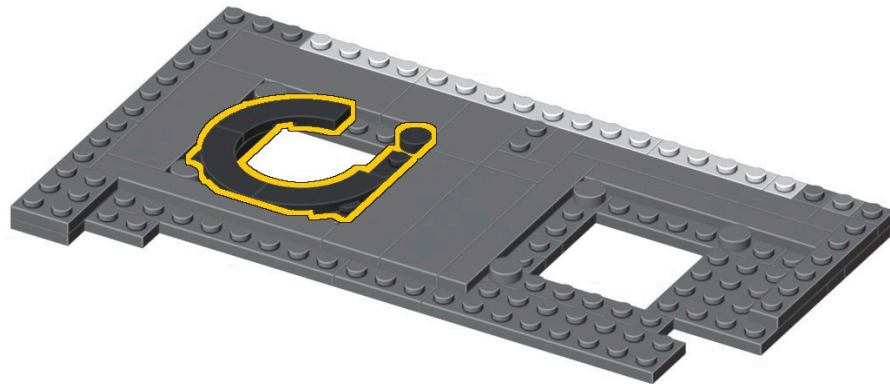
Bottom half



13



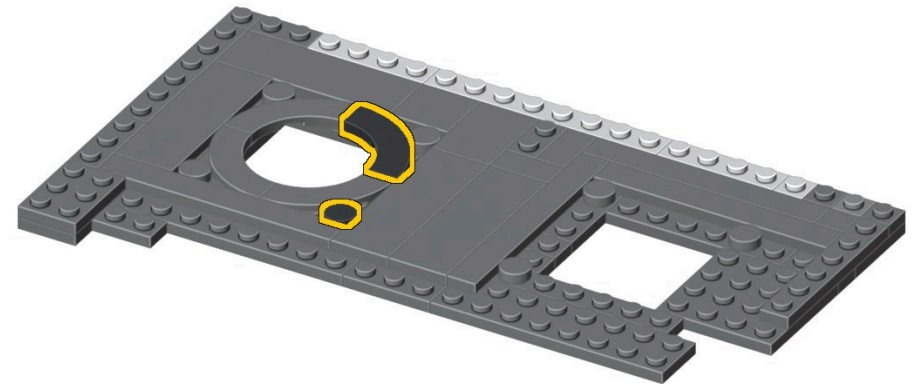
Round rims around the two reel drive holes



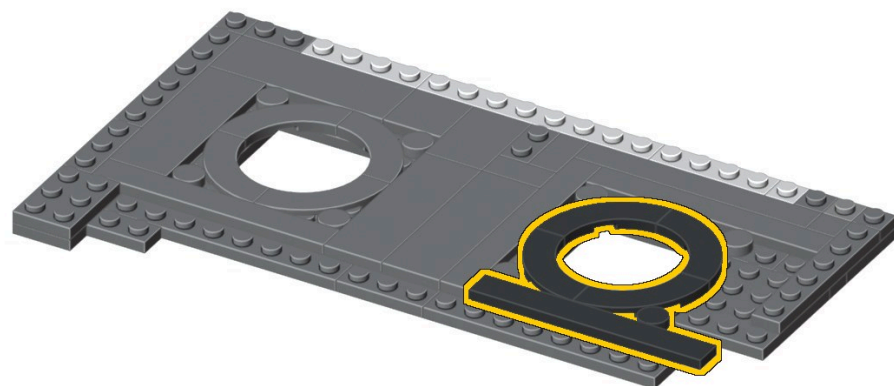
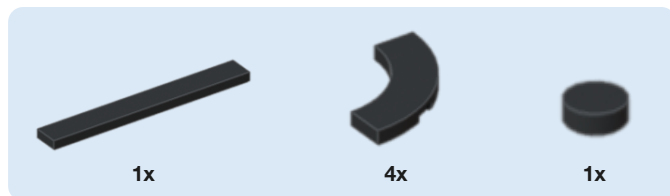
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Bottom half

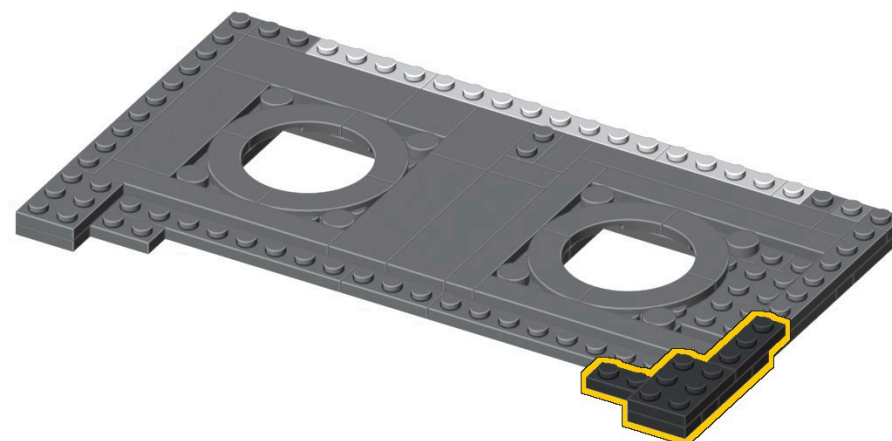
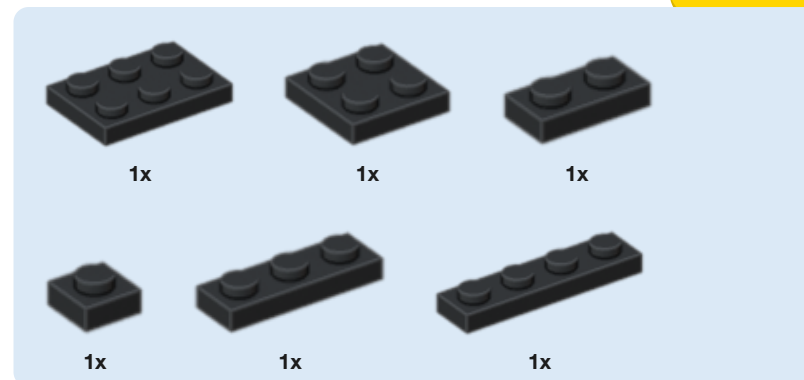


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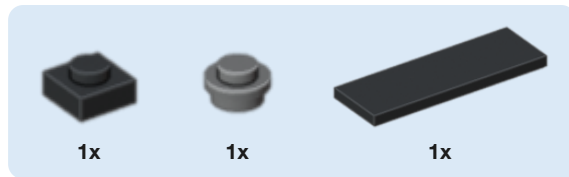


16

Bottom half

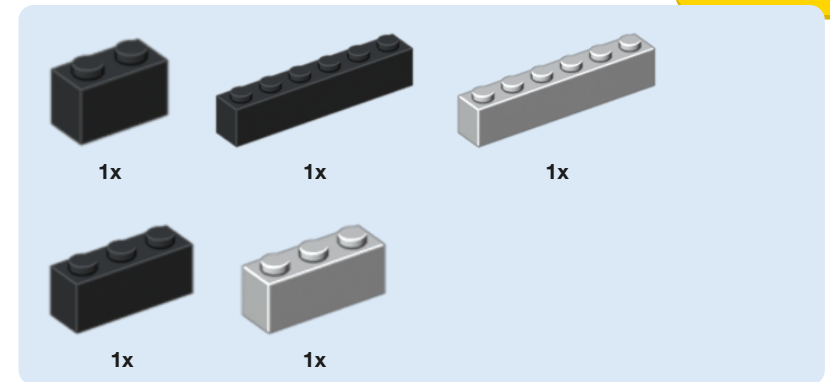


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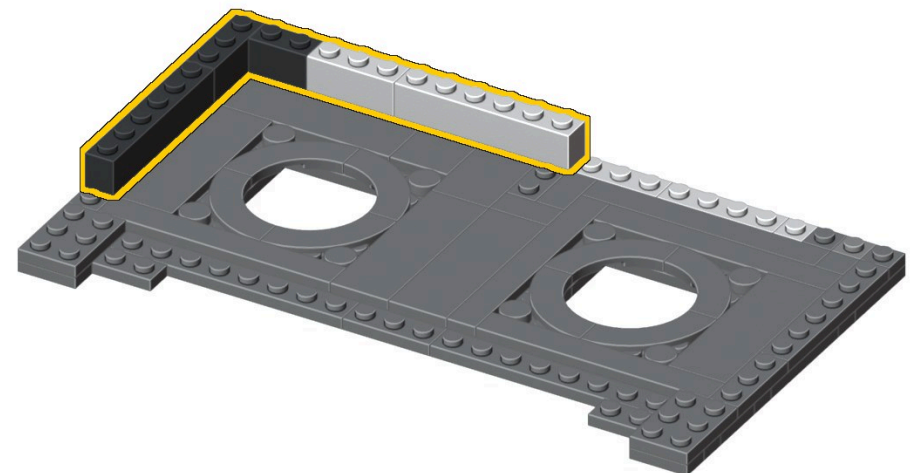


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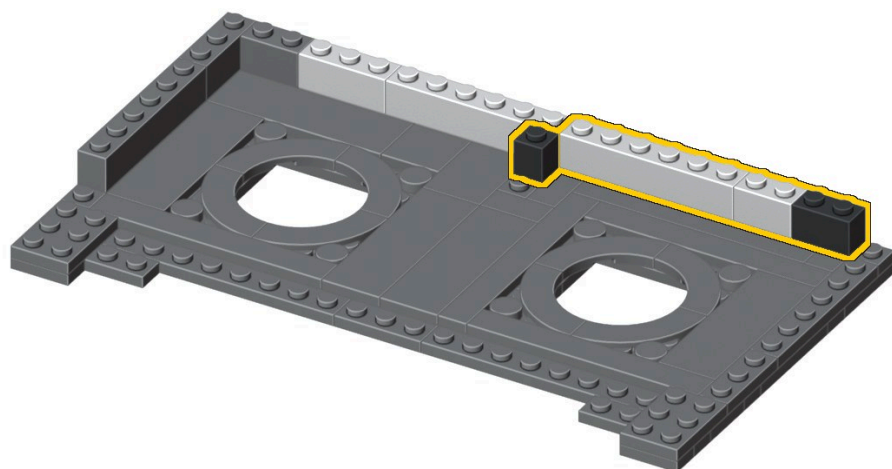
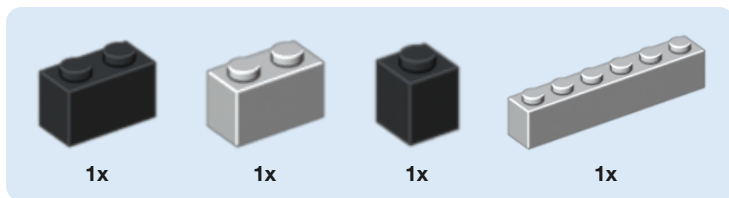
Bottom half



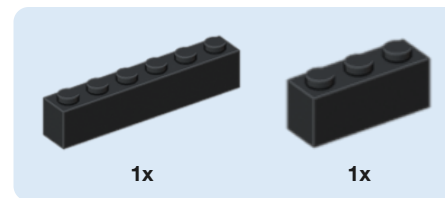
Walls



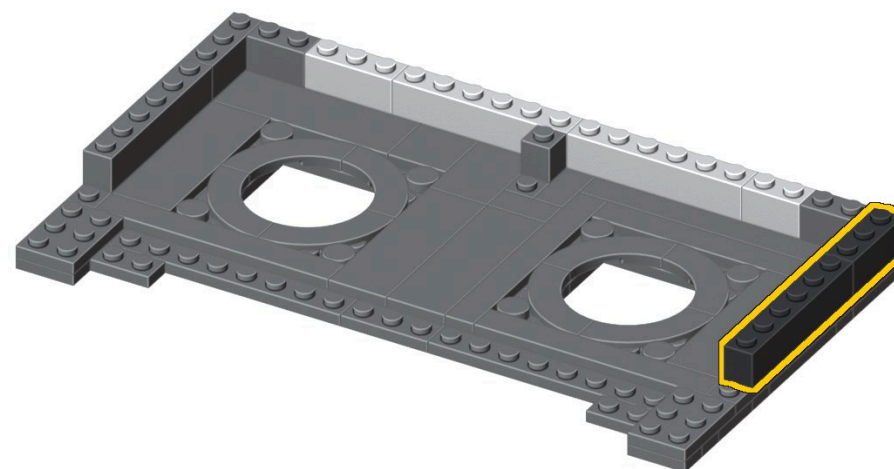
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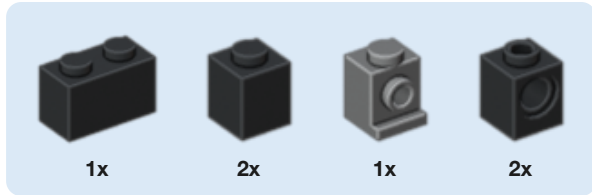
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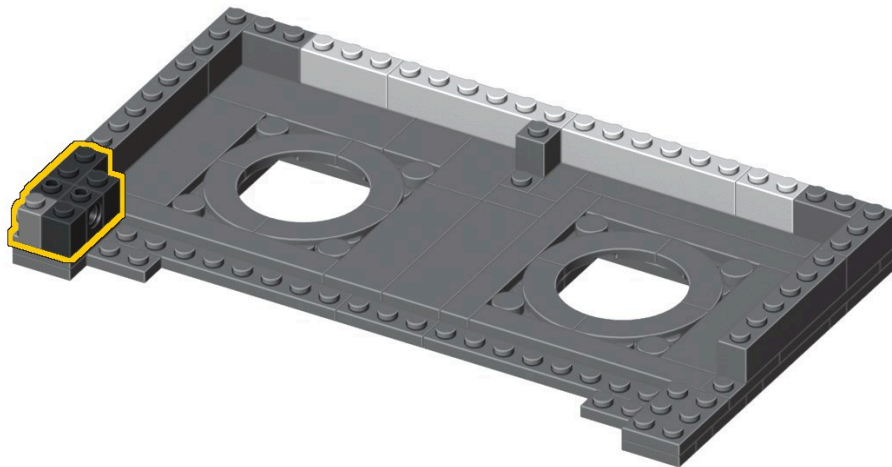
Bottom half



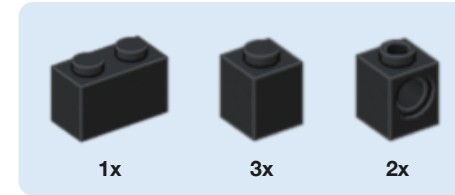
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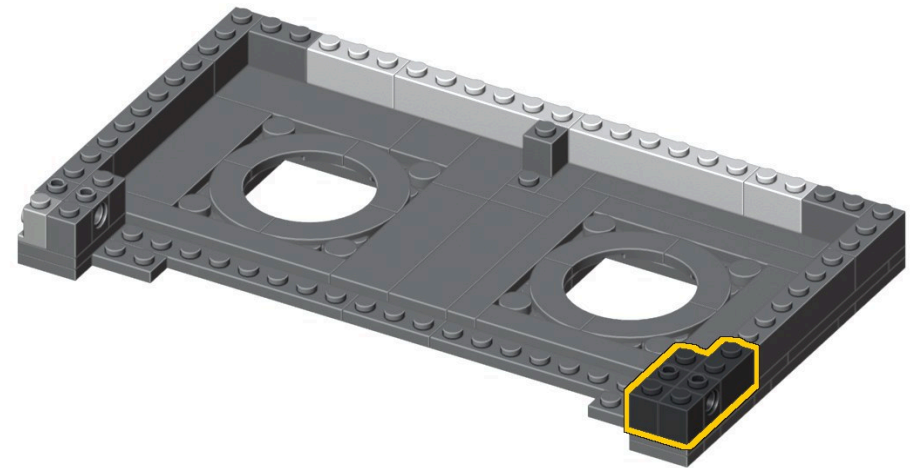
Front corners: a hole through each side for the tape-end light path, and the grey door release button on the right



22



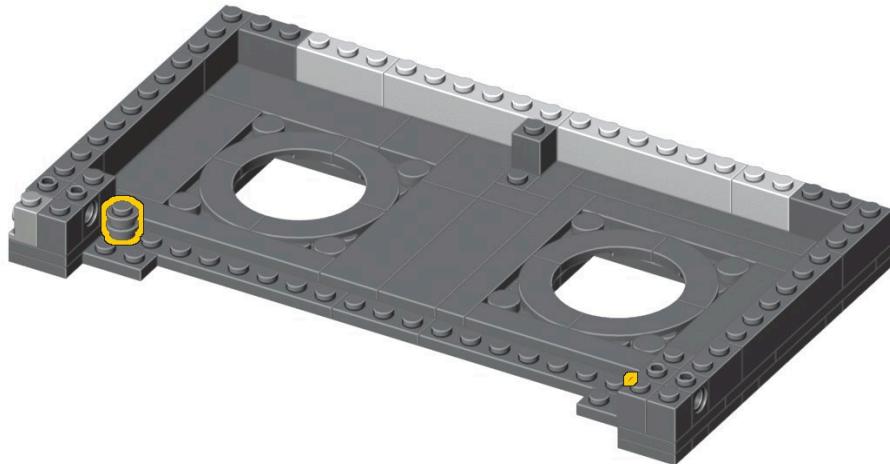
Bottom half



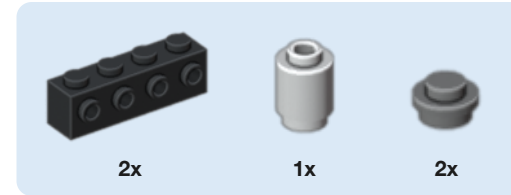
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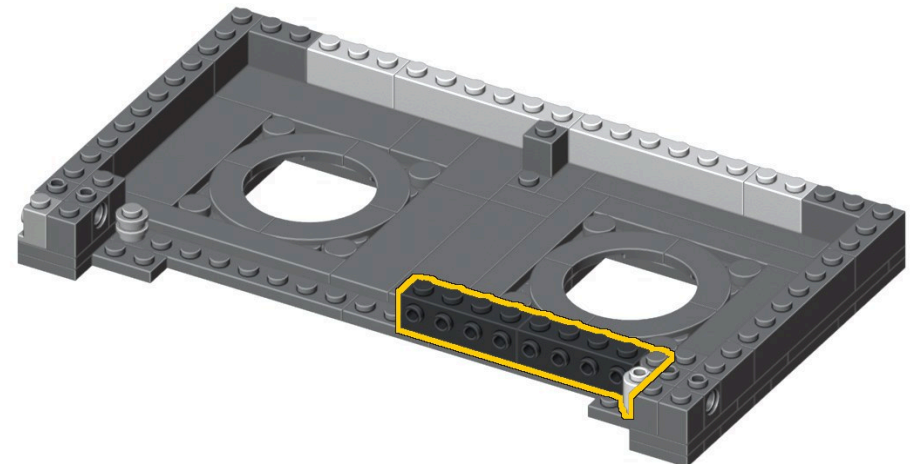
Tape path: backing bricks with side studs, silver guide pins, white rollers



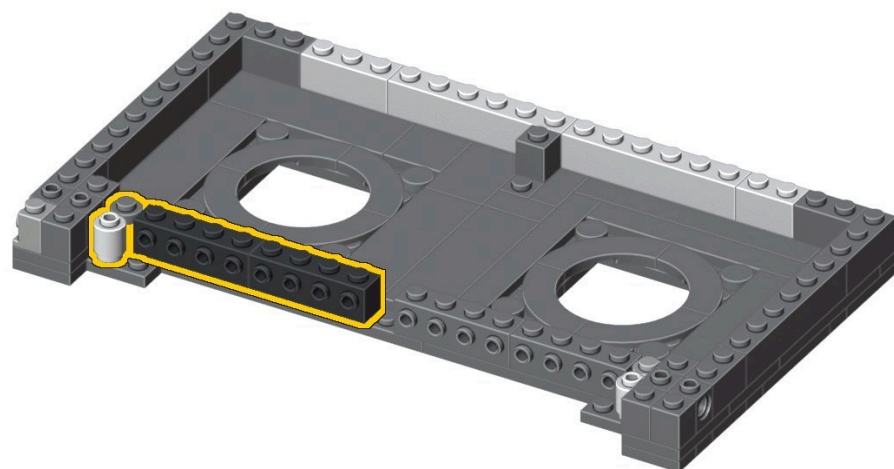
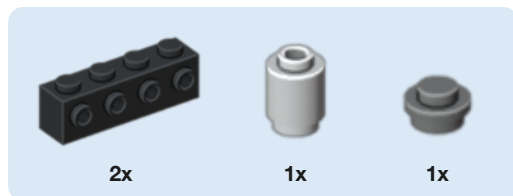
24



Bottom half

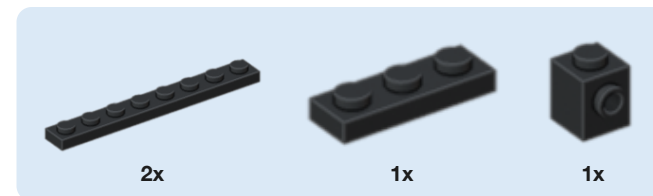


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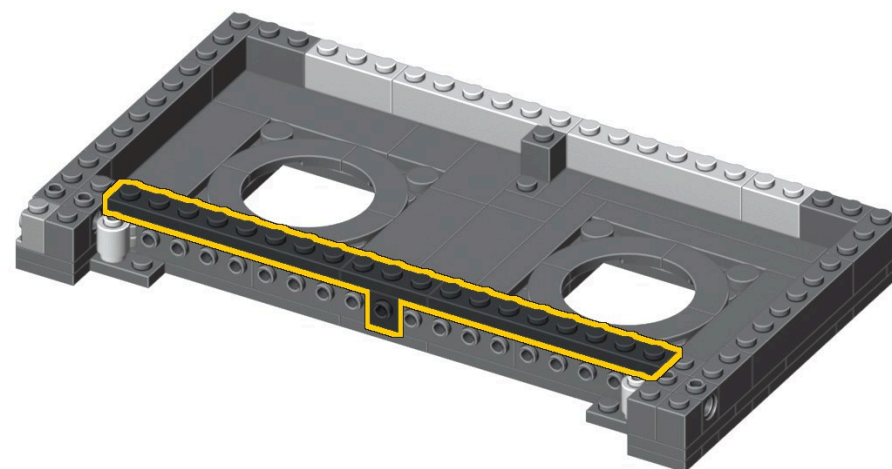


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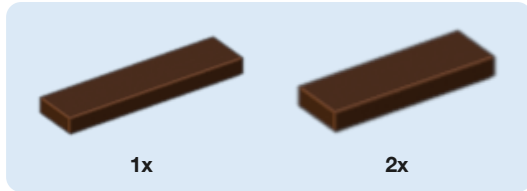
Bottom half



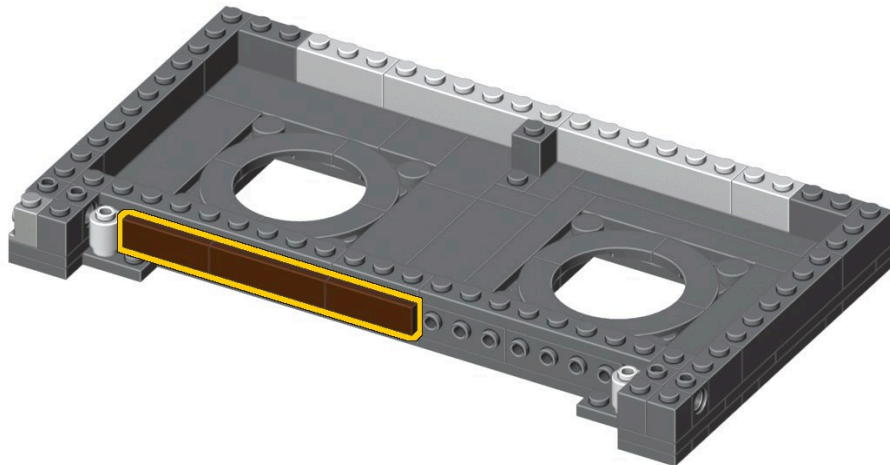
Cap the front beam



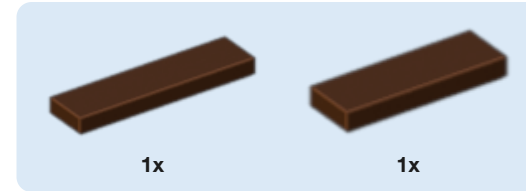
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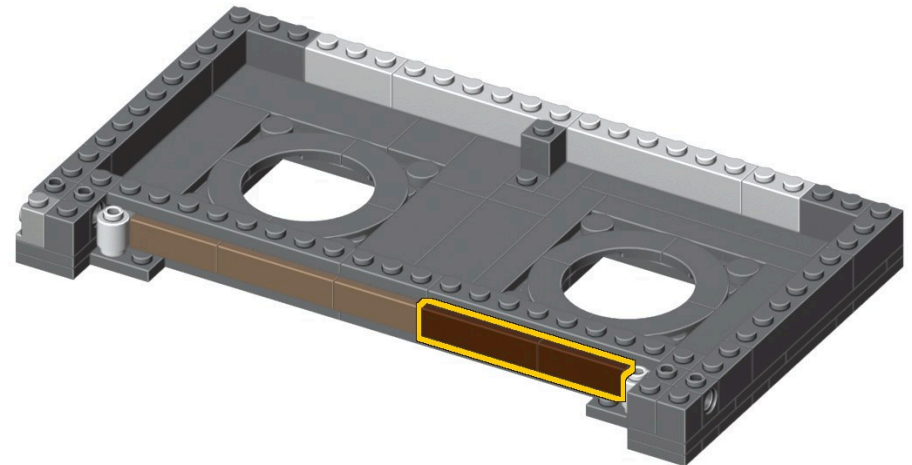
Stretch the tape across the mouth



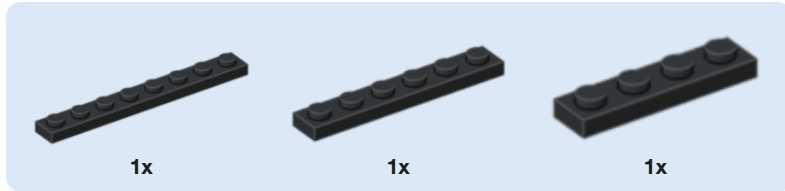
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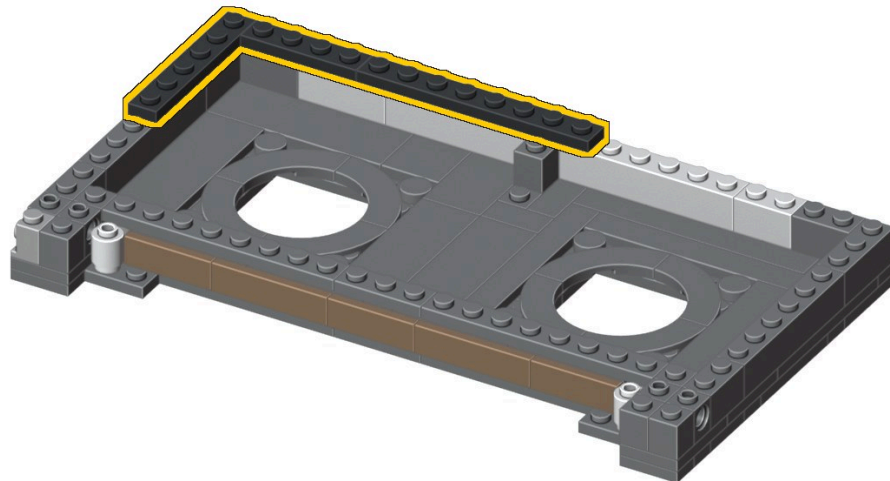
Bottom half



29

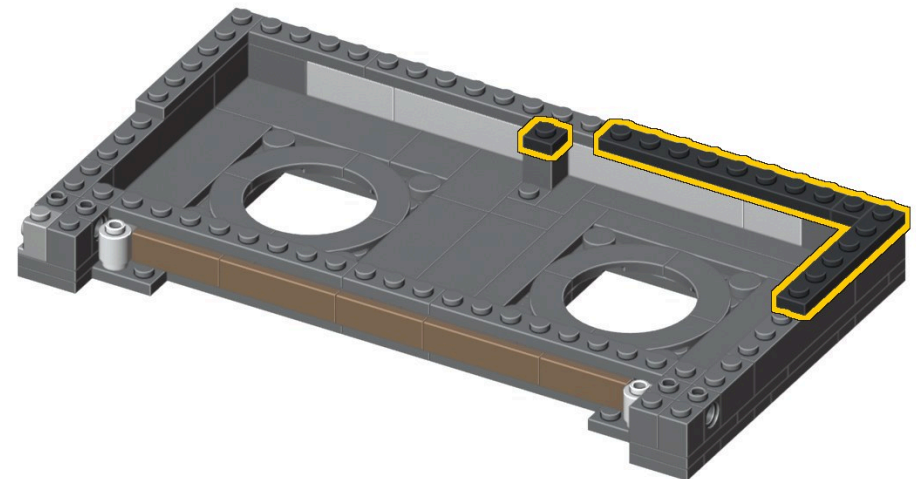
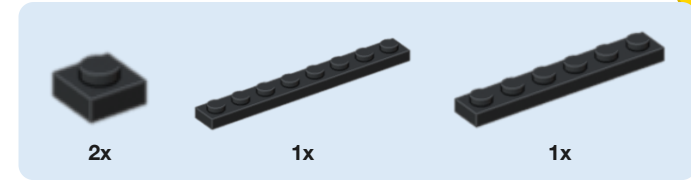


Top course: ties the walls, corners and front beam together

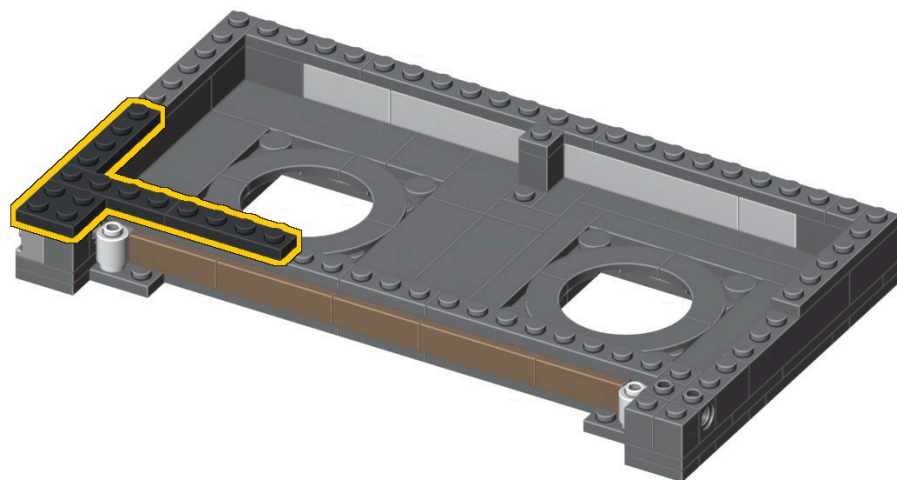
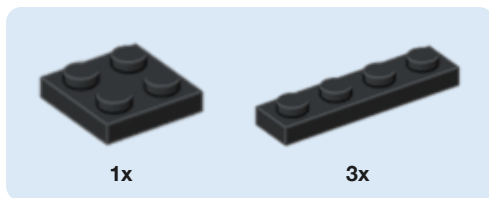


30

Bottom half

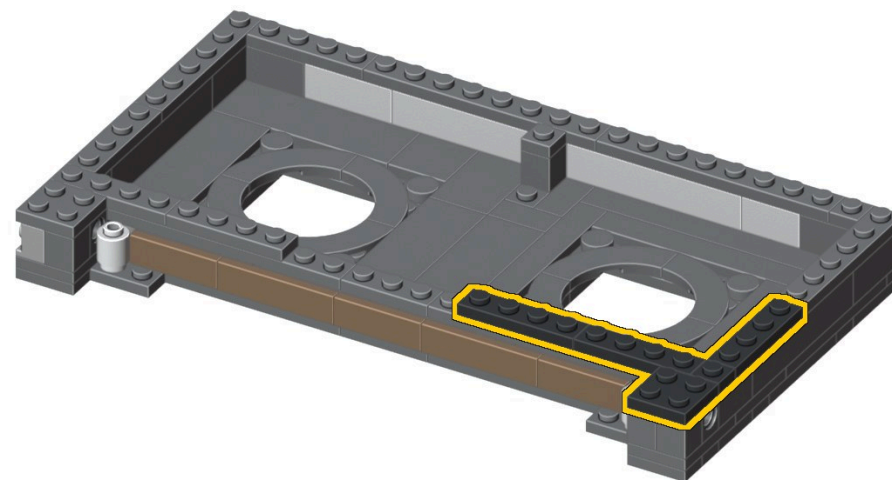
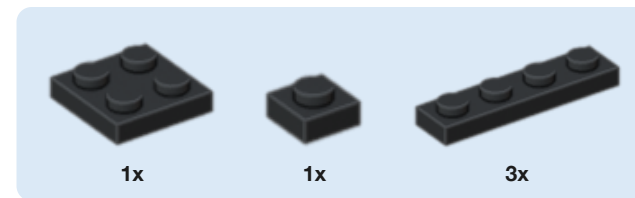


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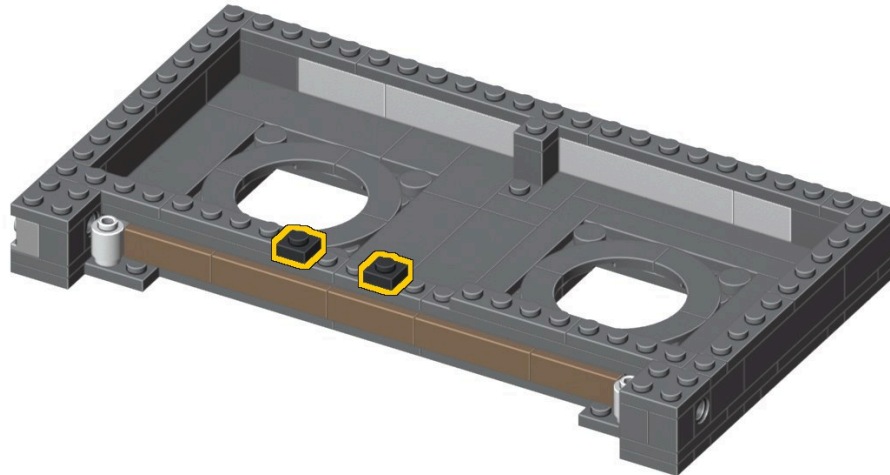
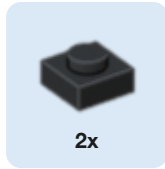


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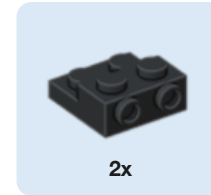
Bottom half



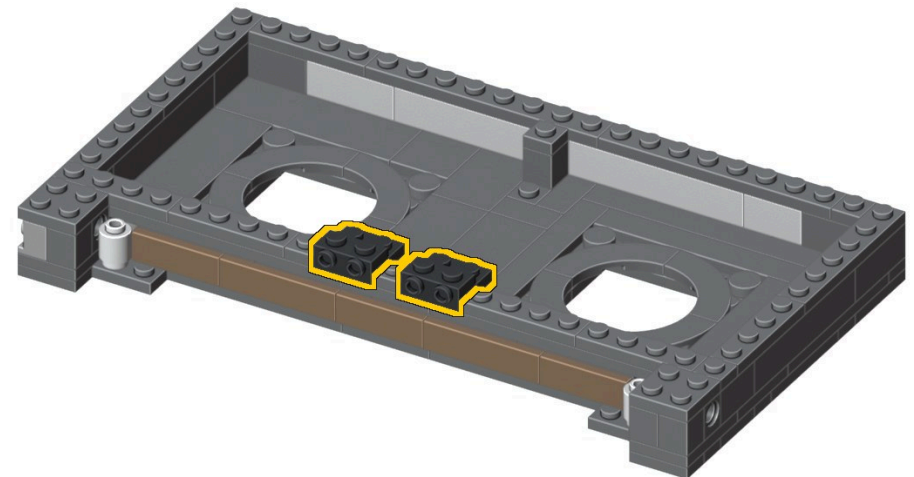
33



34

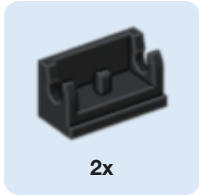


Door hinge mounts: plates with studs on the side

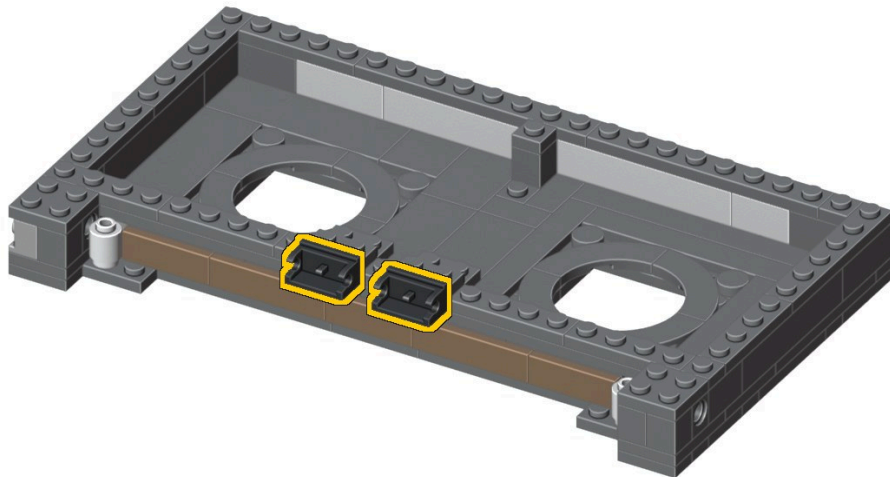


Bottom half

35

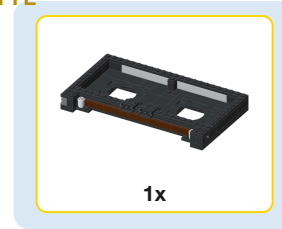


Hinge bases, pushed onto the side studs

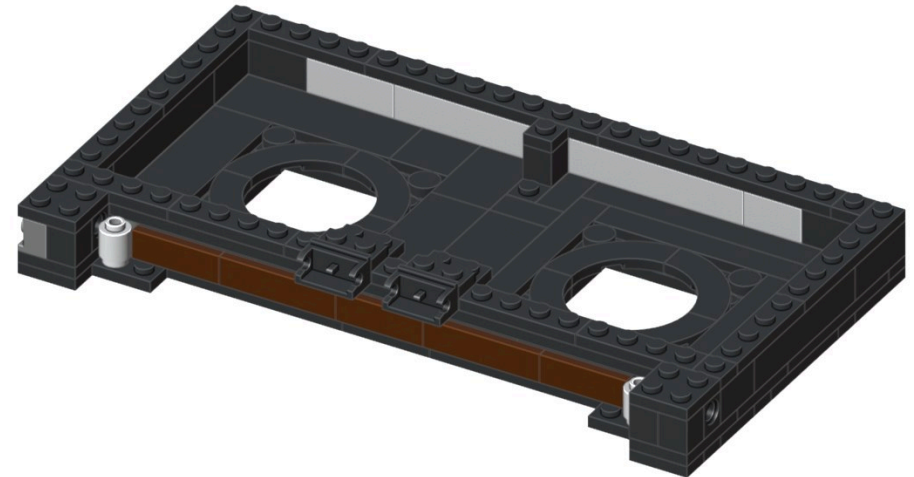


VHS CASSETTE

36



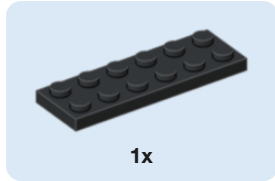
The bottom half



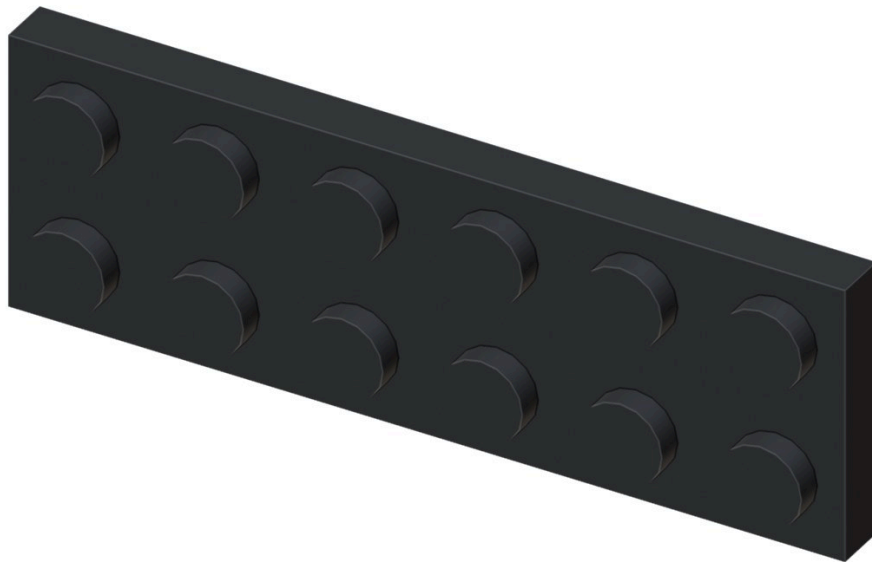
VHS Cassette

DUST DOOR

37



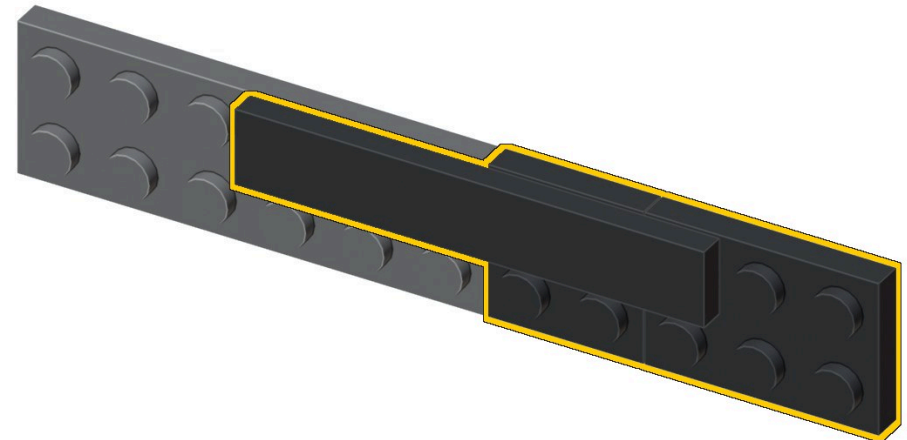
Dust door, built flat: start with a 2 x 6 plate



38

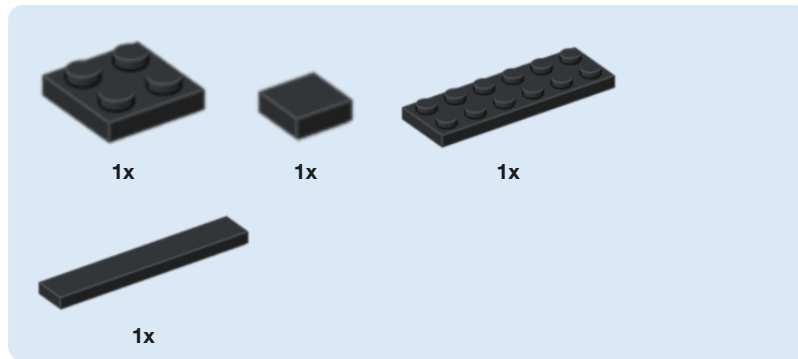


Add plates and tie them with a tile

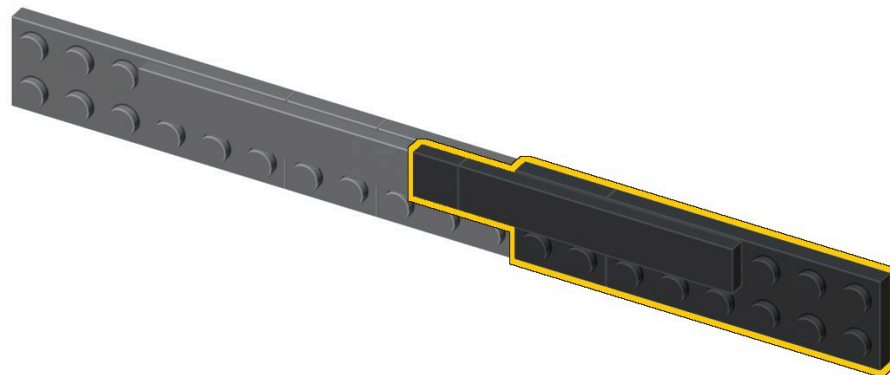


Dust door

39

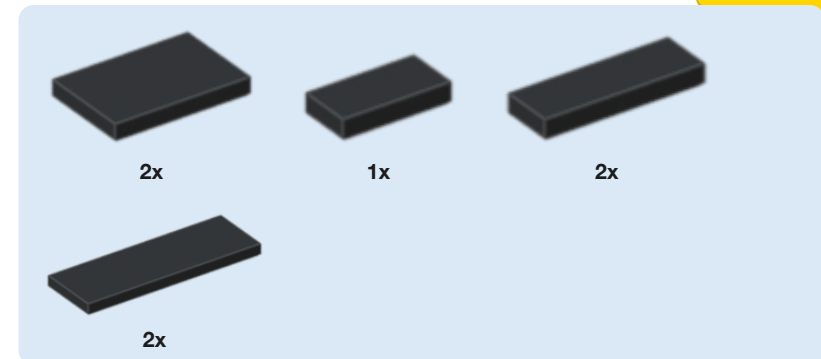


Complete the plate row

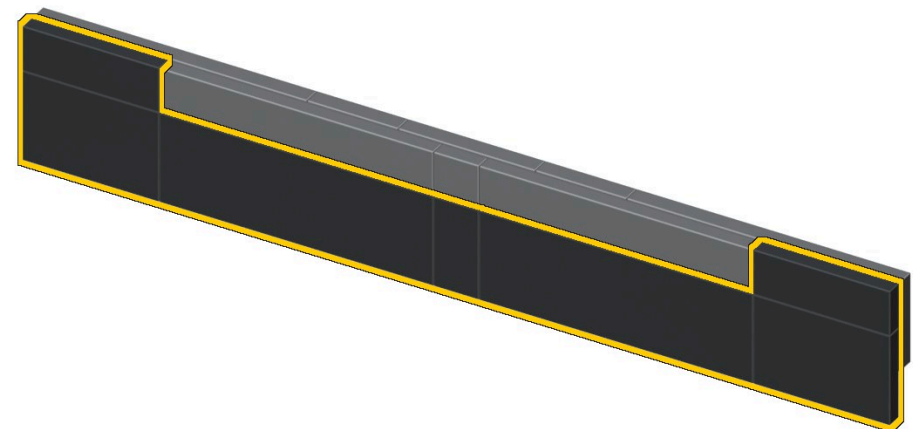


40

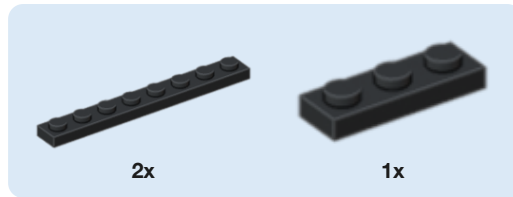
Dust door



Smooth outer face

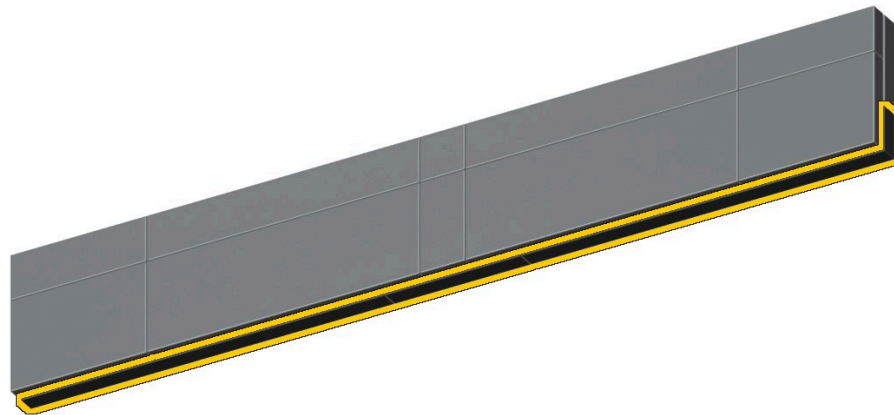


41

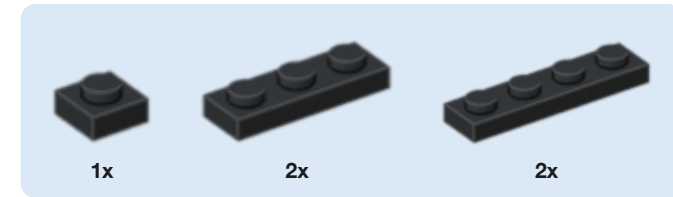


Lower edge of the door

seen from below

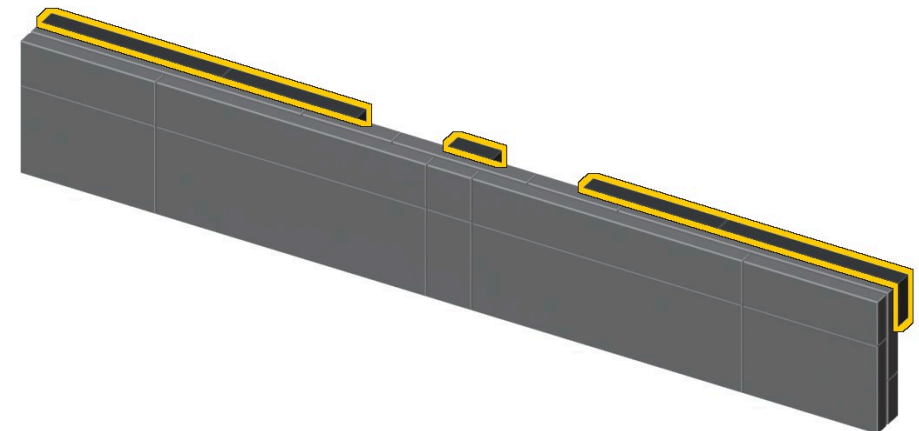


42

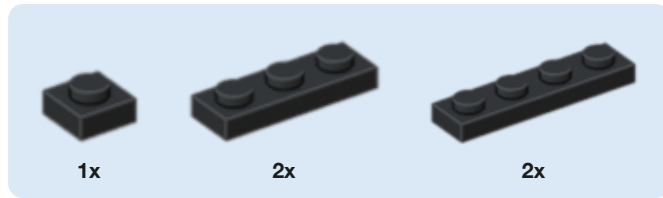


Top edge: fillers that swing up with the door

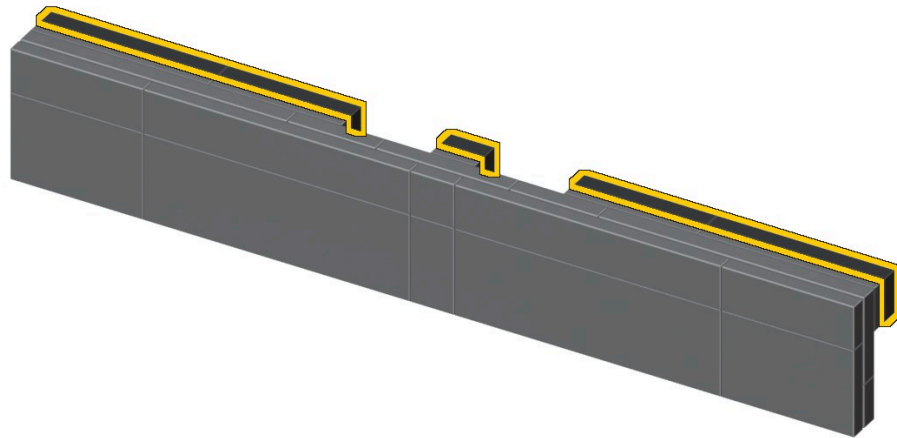
Dust door



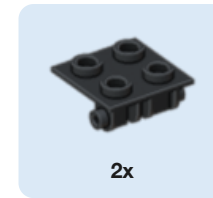
43



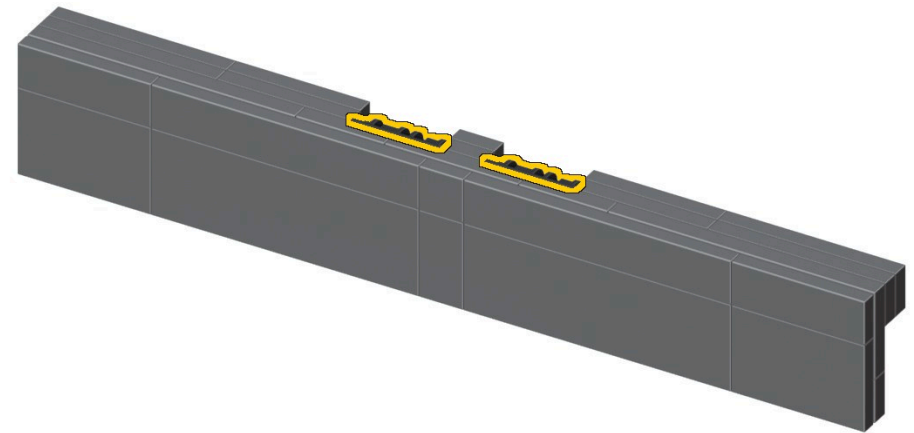
Second layer of fillers



44



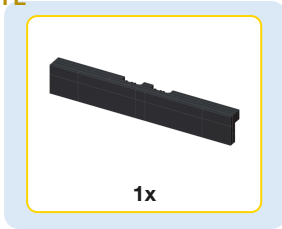
Hinge tops



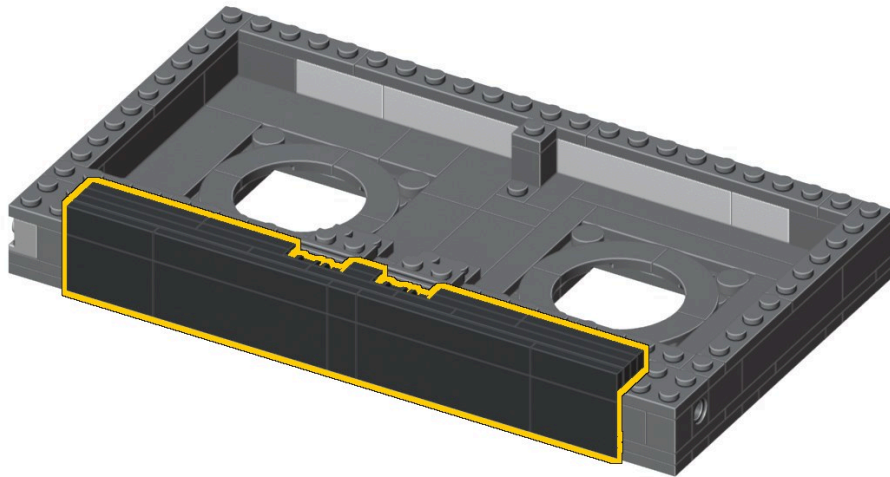
Dust door

VHS CASSETTE

45



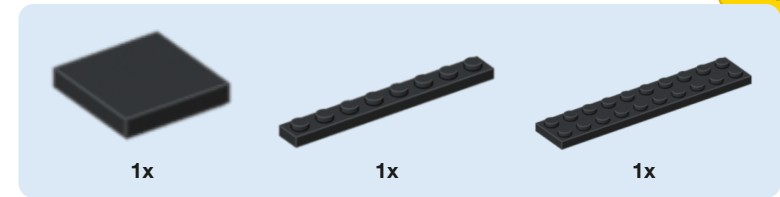
Click the dust door onto its two hinges



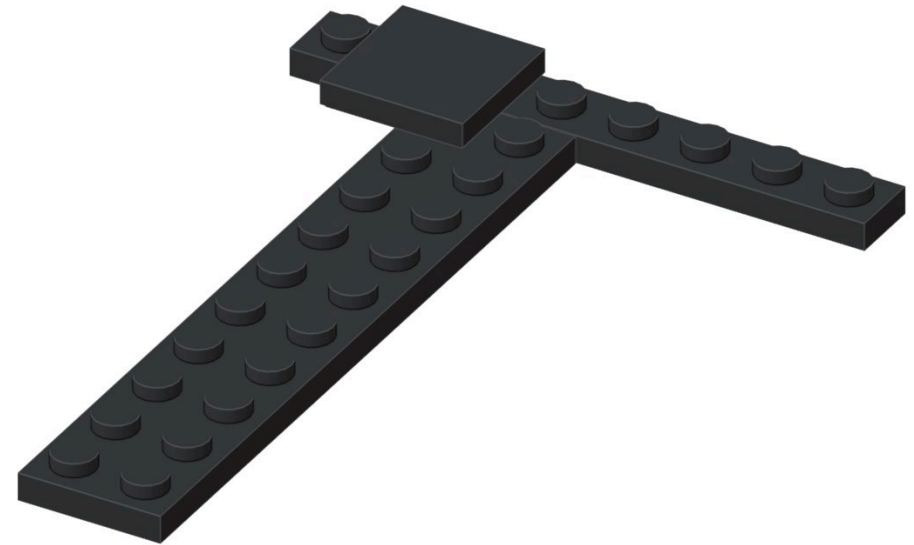
TOP HALF

46

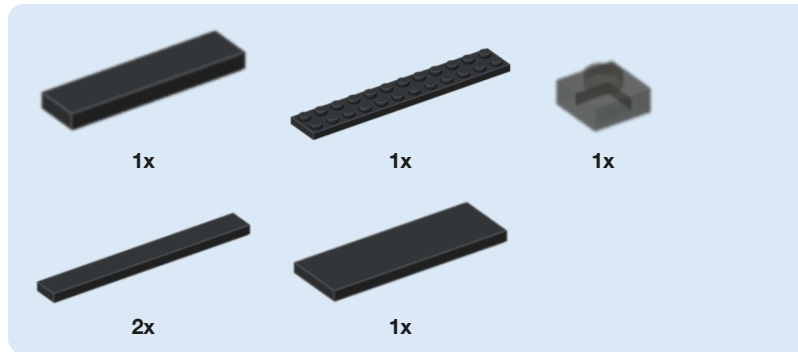
Top half



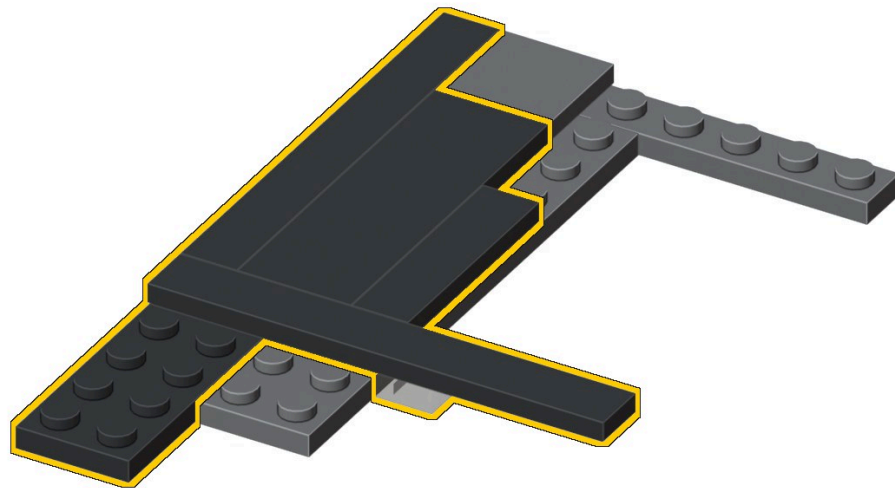
Top half, built face up: plates first



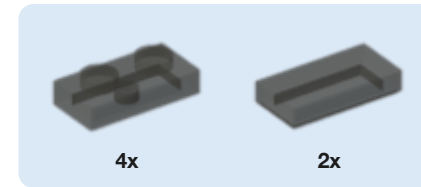
47



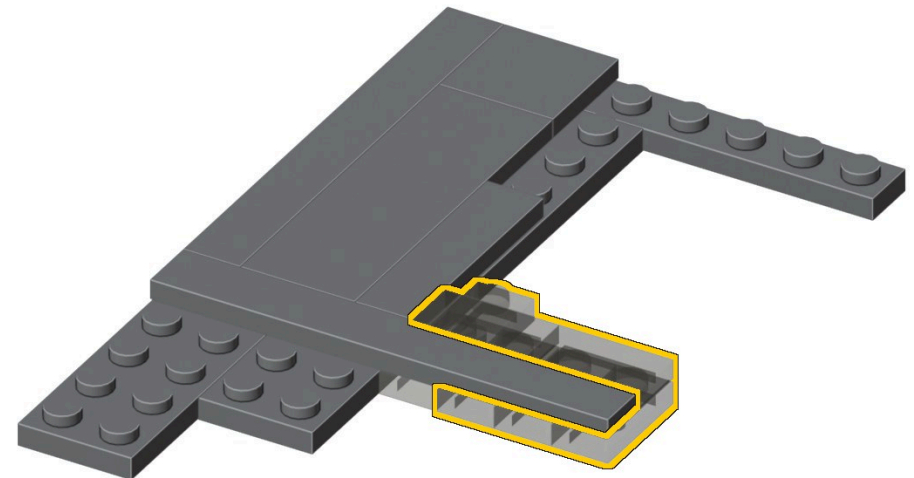
Smoked window: see-through plates, and black holders over the reel hubs



48

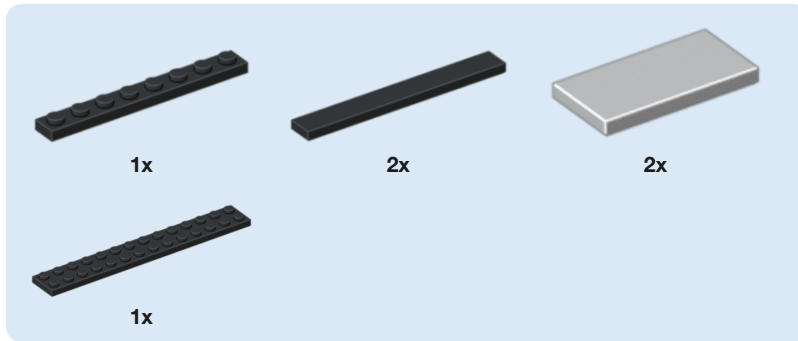


Window glass tiles

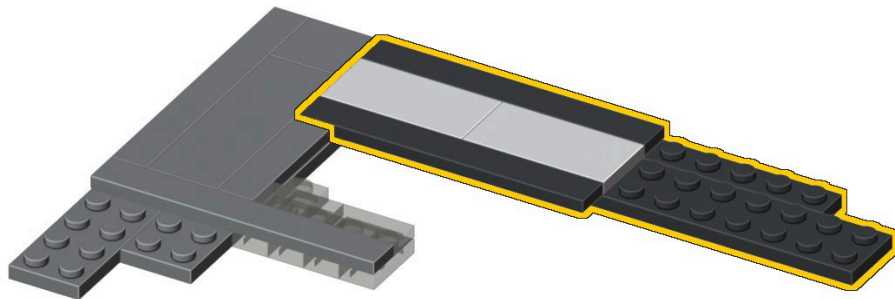


Top half

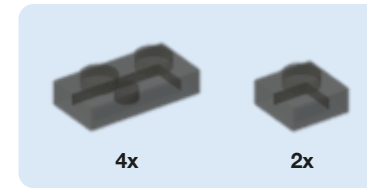
49



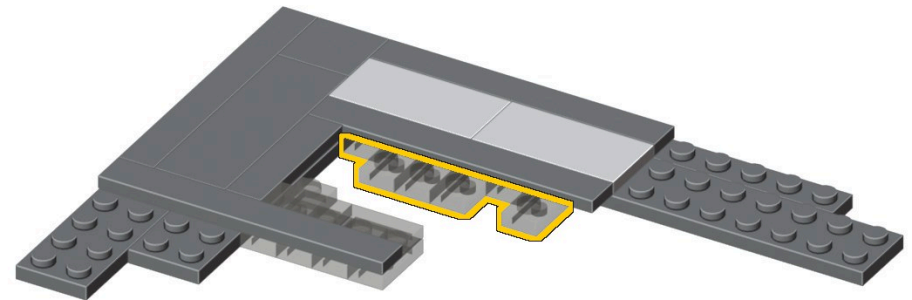
White face label recess



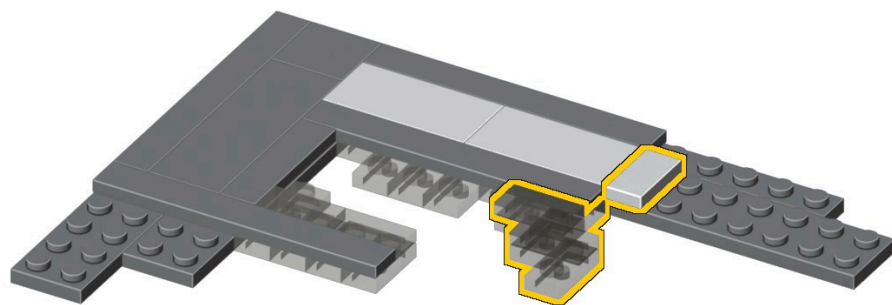
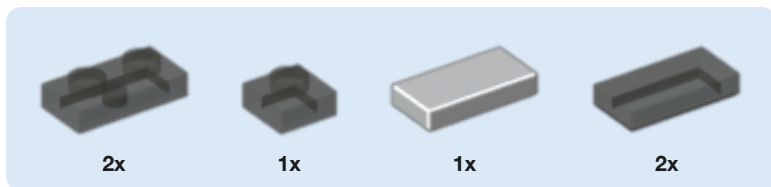
50



Top half

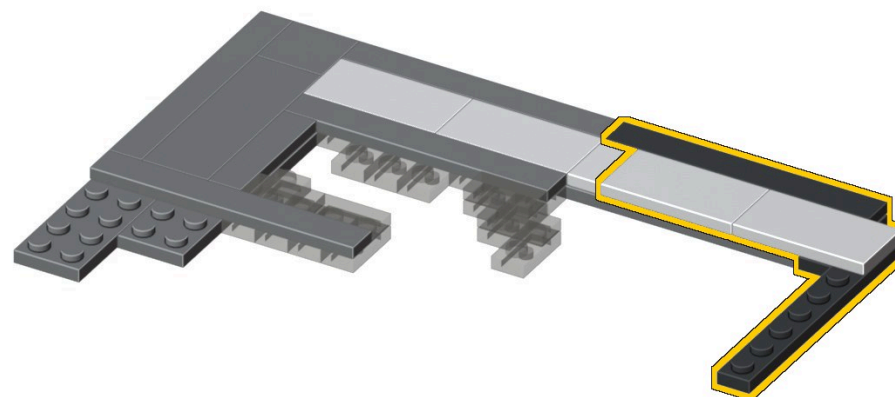
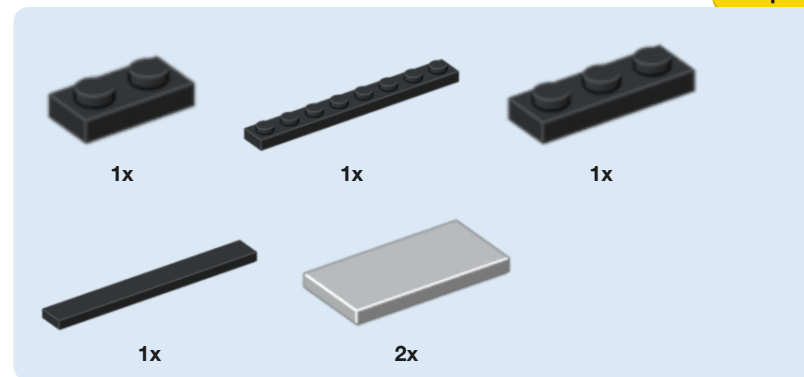


51

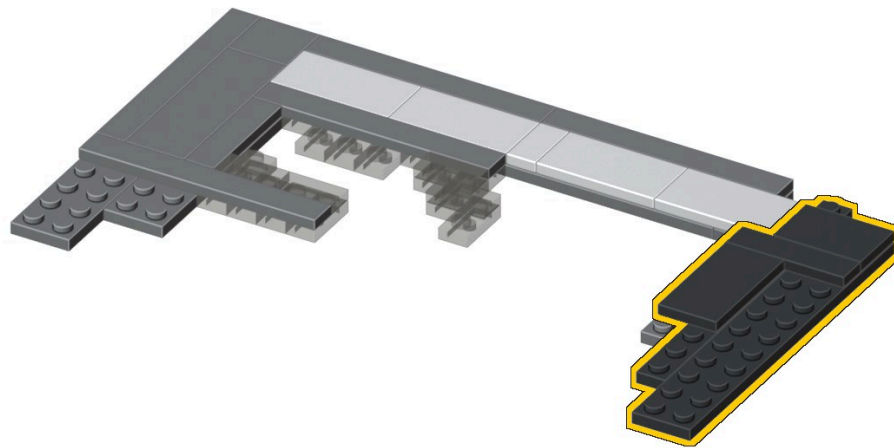
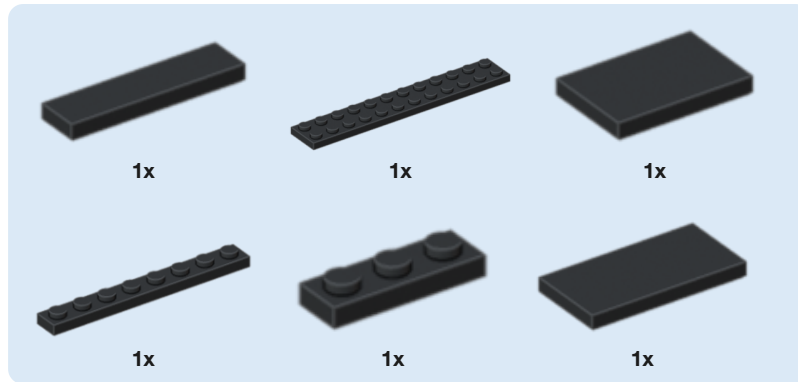


52

Top half

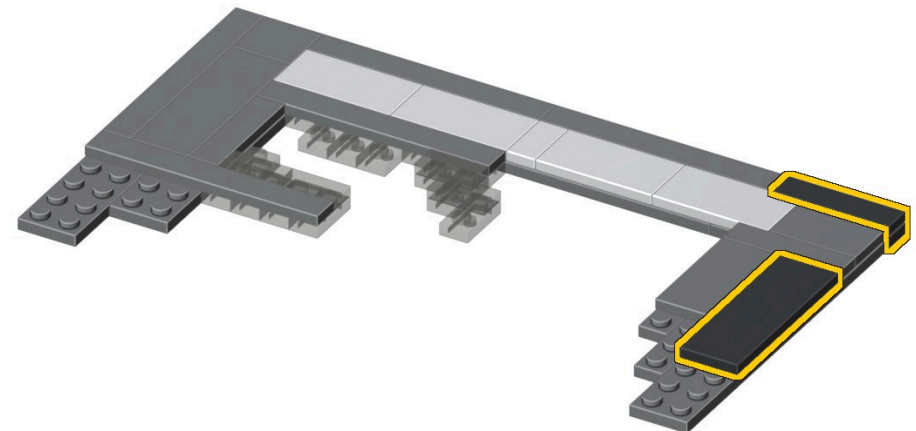
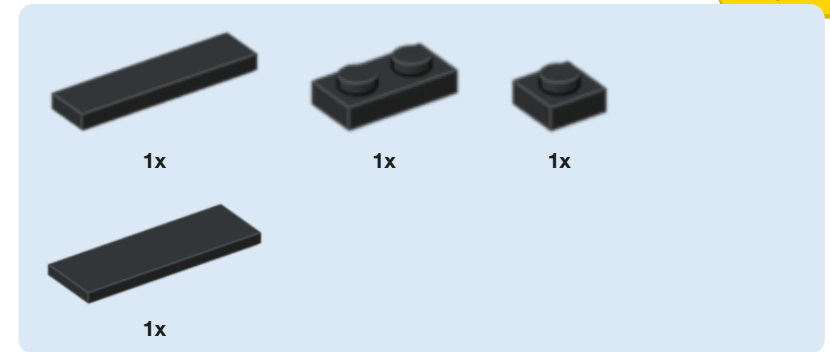


53

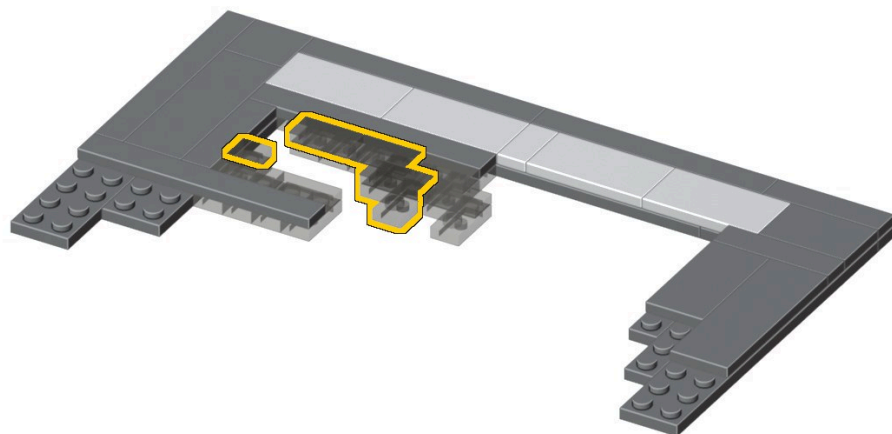
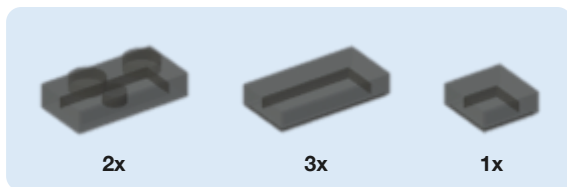


54

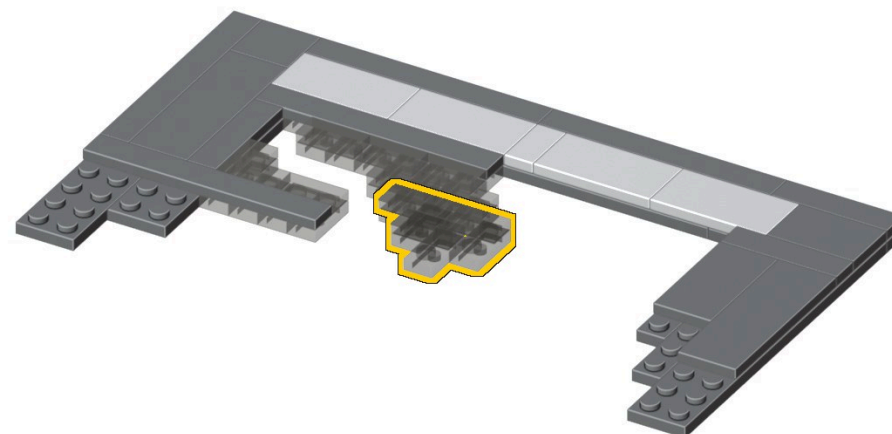
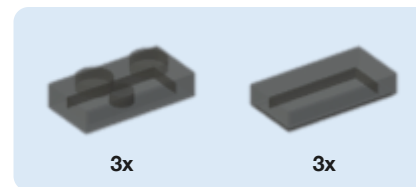
Top half



55

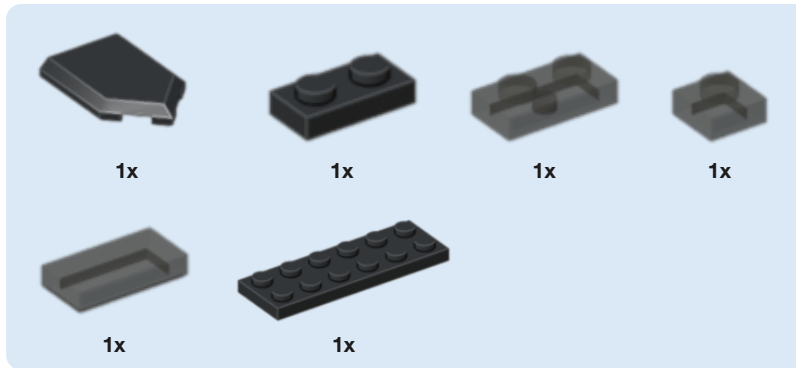


56

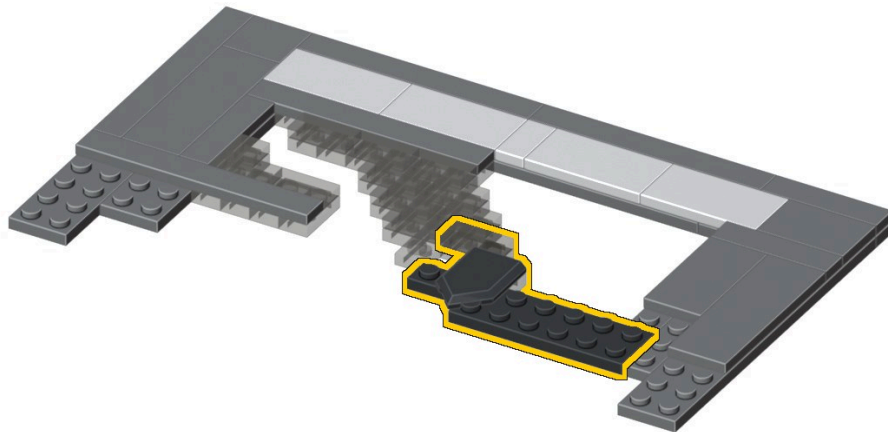


Top half

57

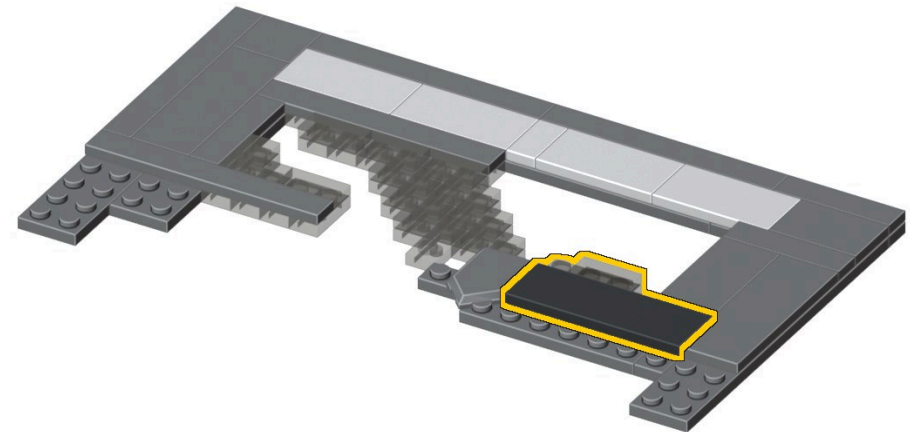
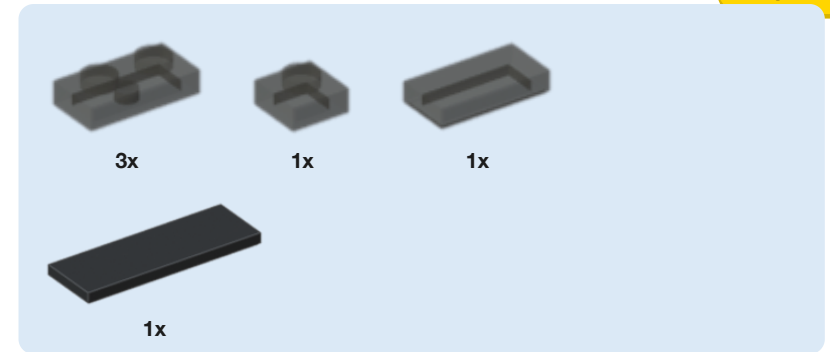


Moulded arrows pointing to the door

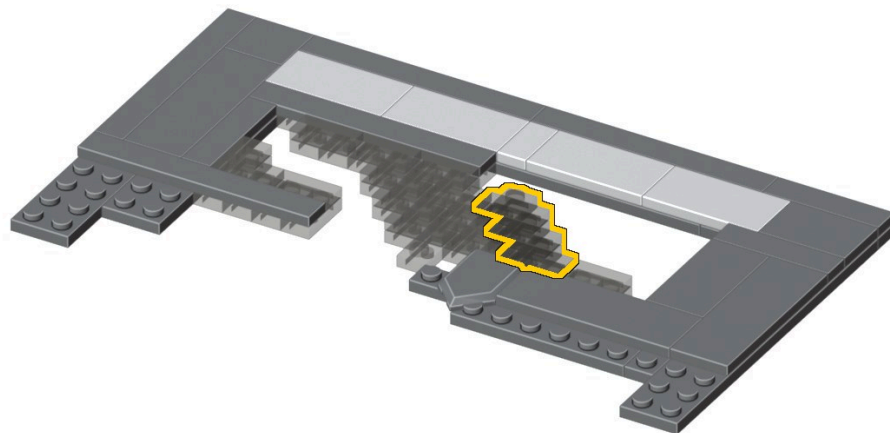
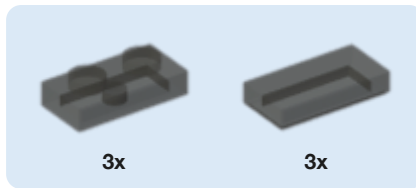


58

Top half

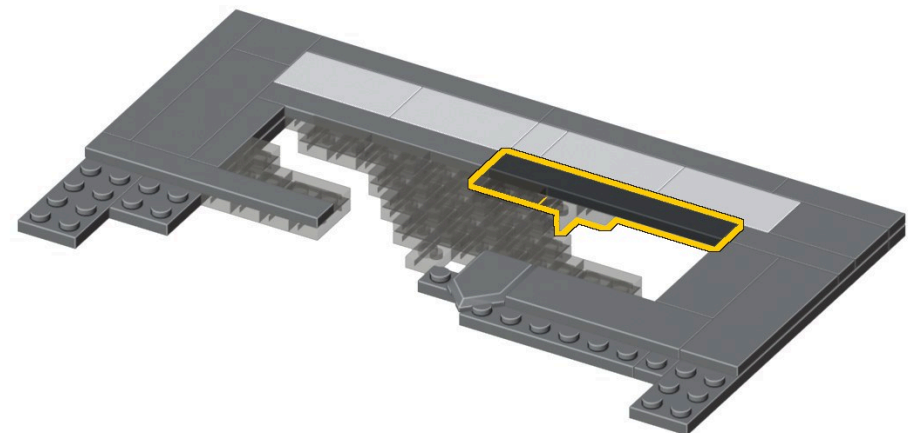
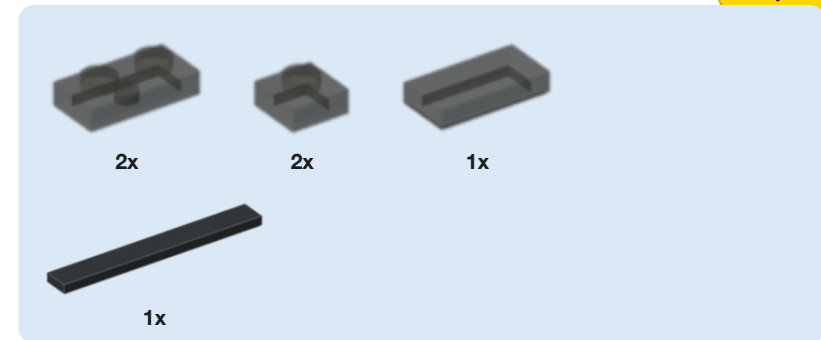


59

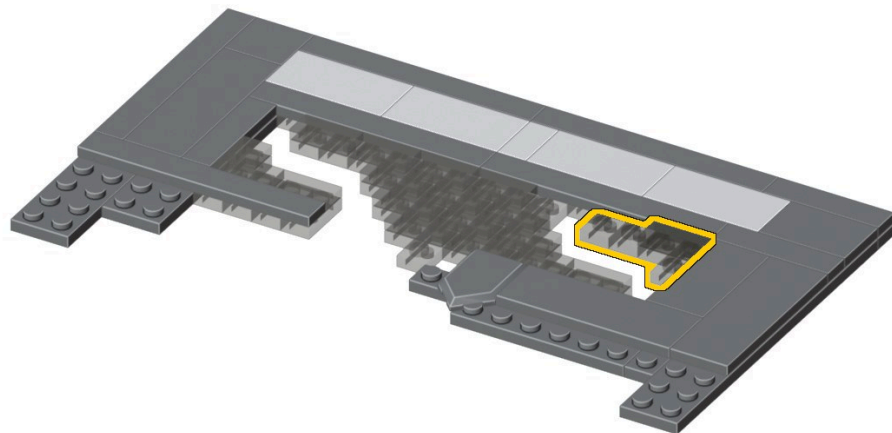
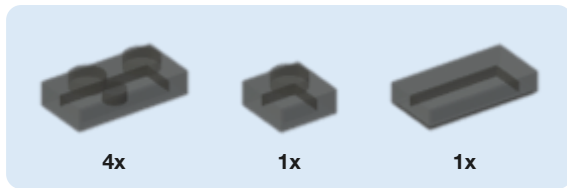


60

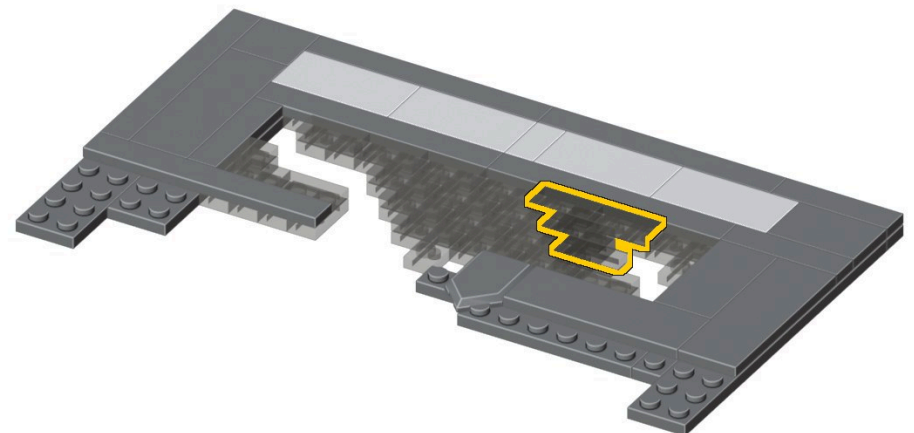
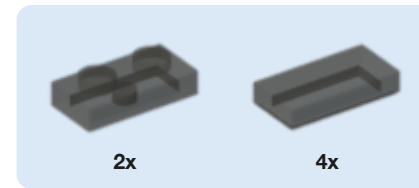
Top half



61

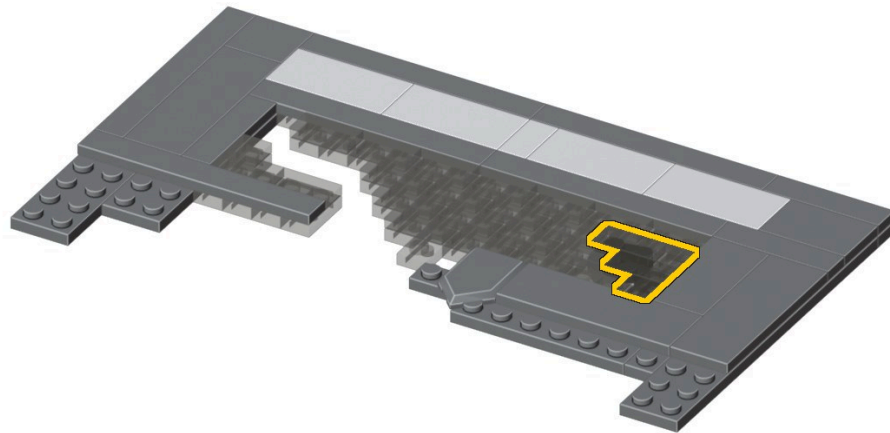
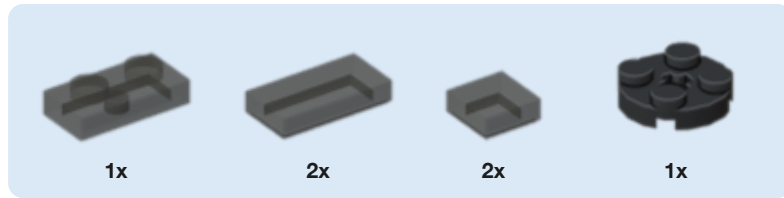


62



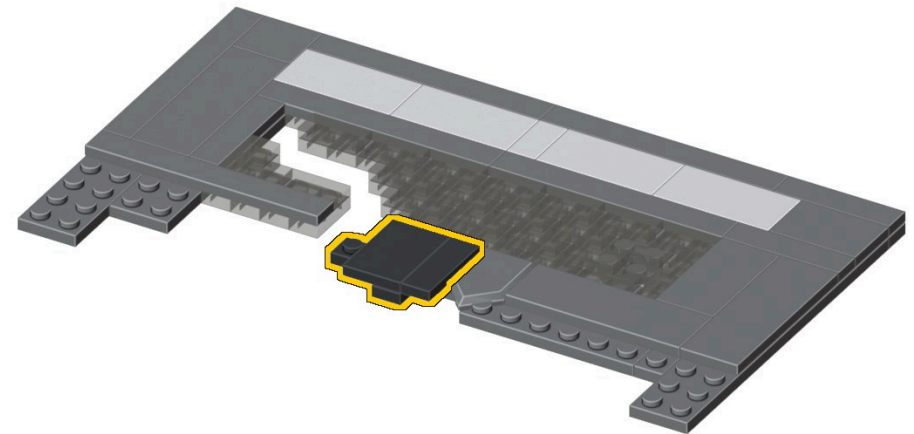
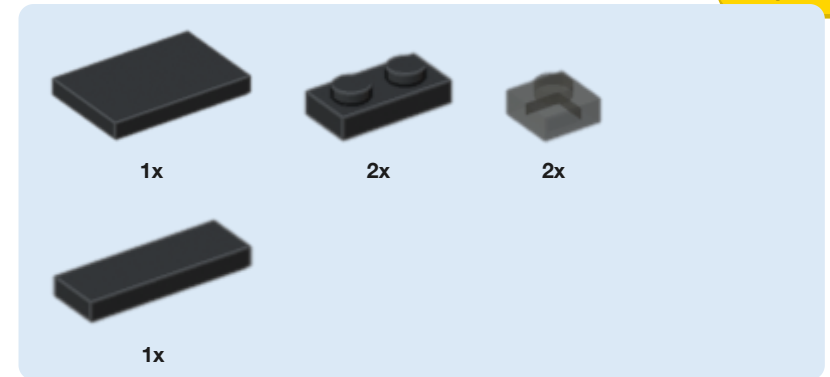
Top half

63

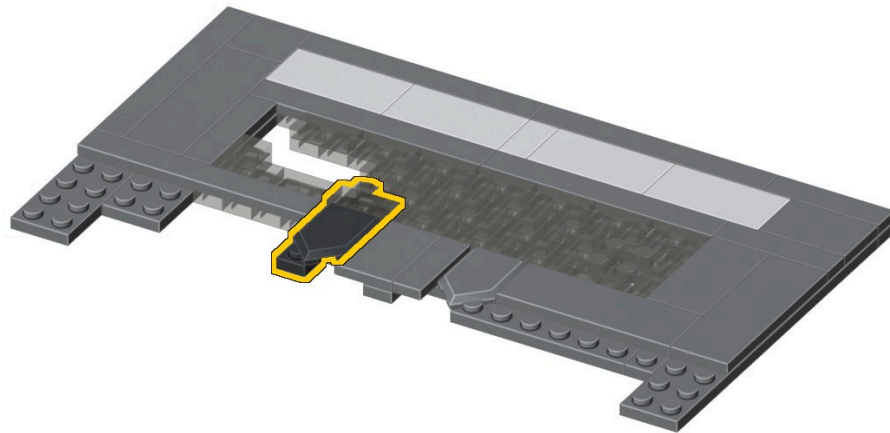
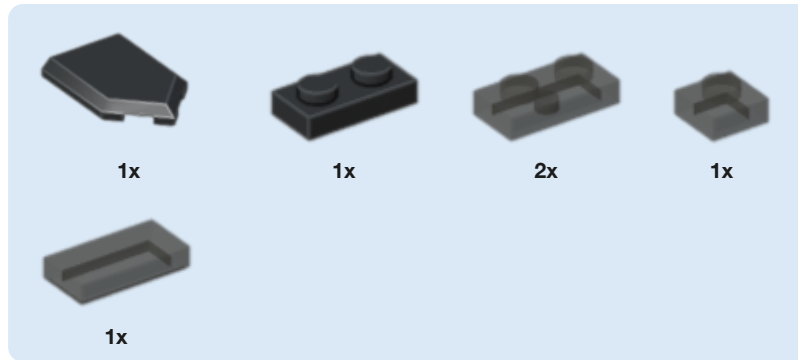


64

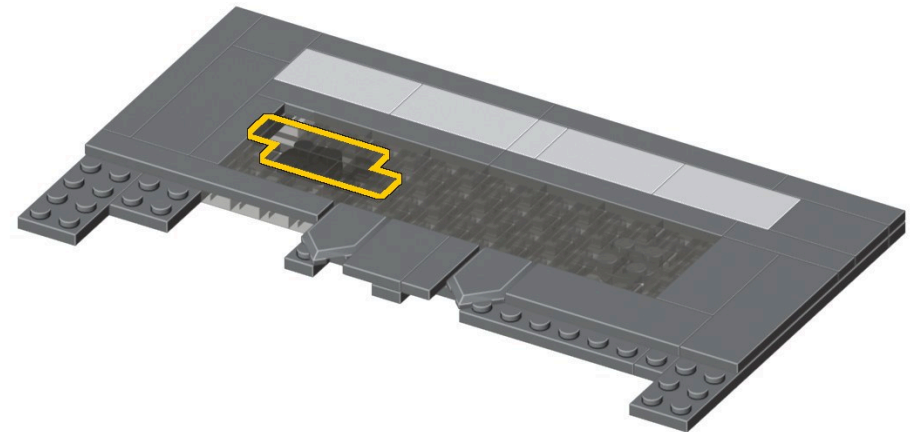
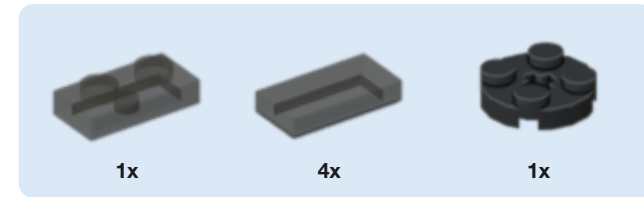
Top half



65

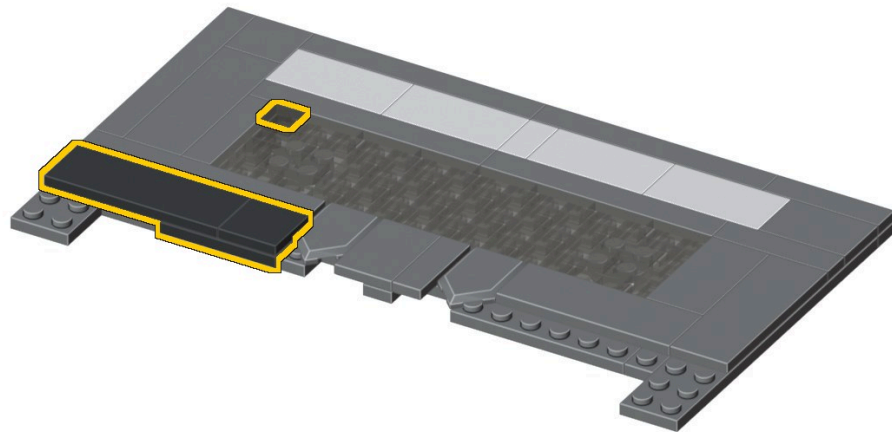
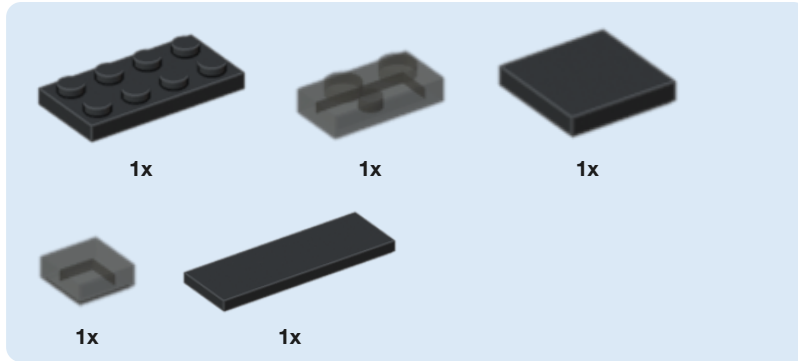


66

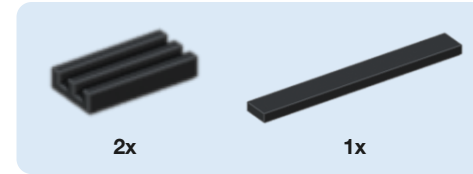


Top half

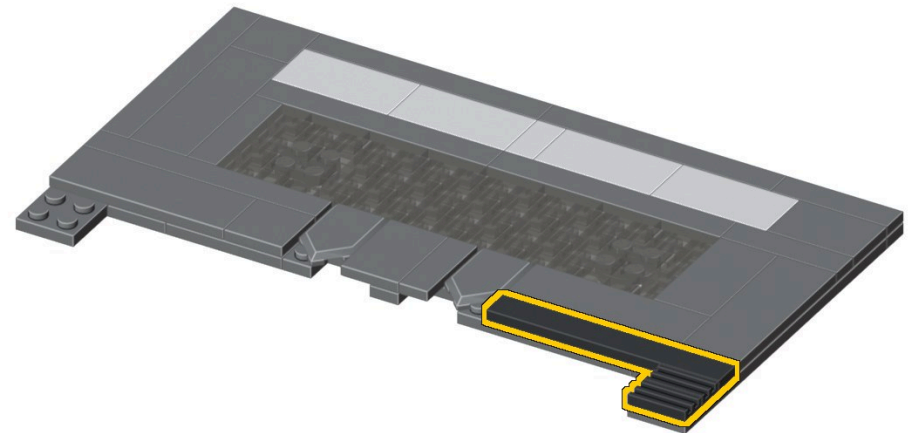
67



68

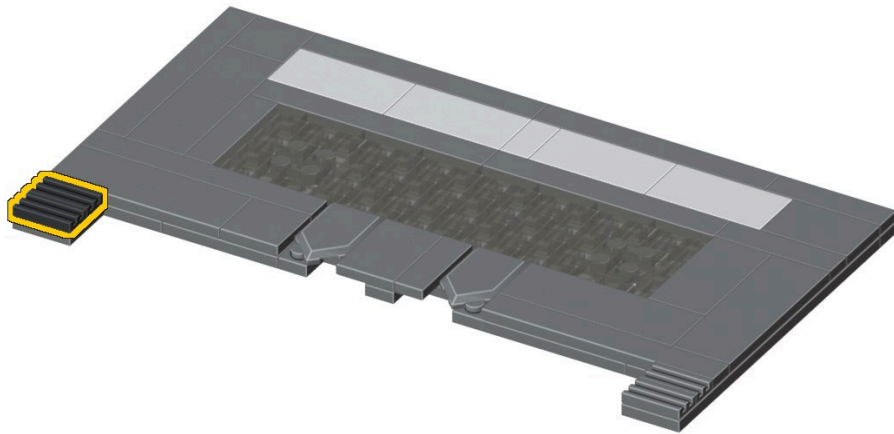
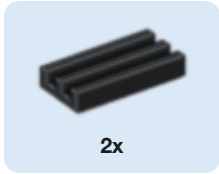


Grip ridges on the front corners



Top half

69

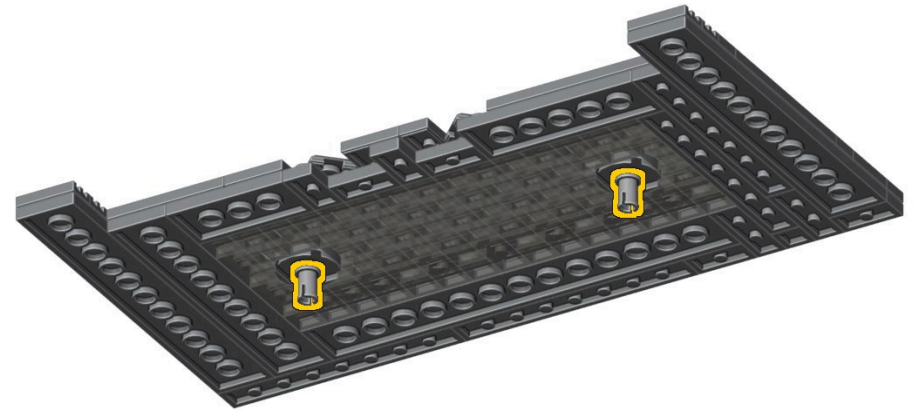


70



Turn the top half over and push in the two reel pins

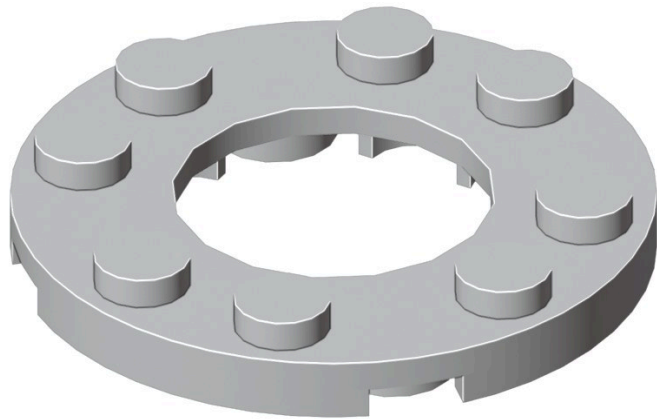
Top half
seen from below



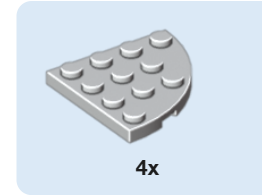
71



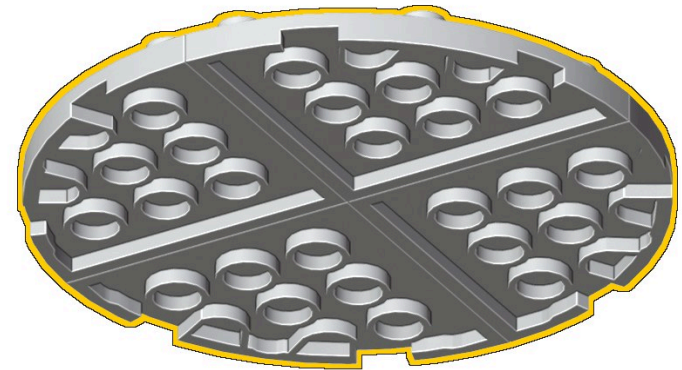
Hub core: the pin runs through its opening



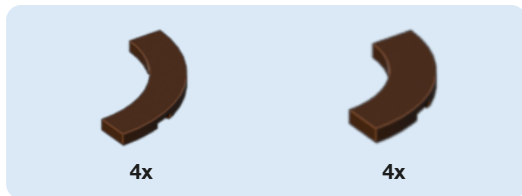
72



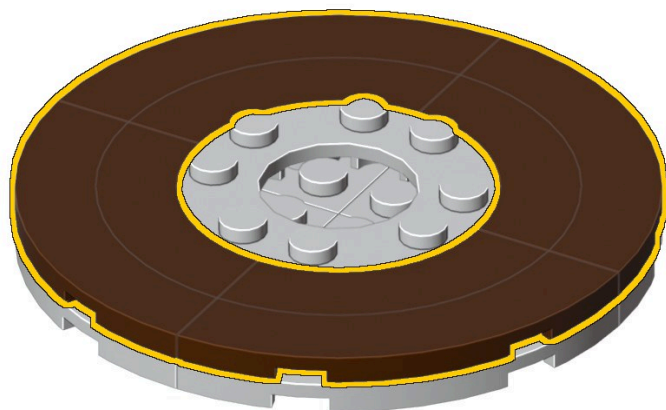
Lower flange from four quarter circles



73



Tape wound on the reel

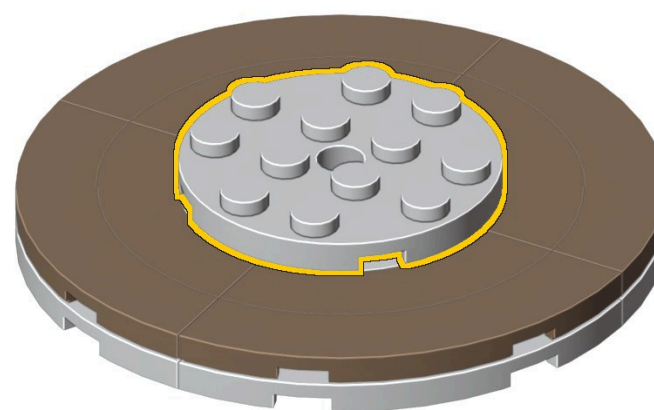


74

Supply reel (full)

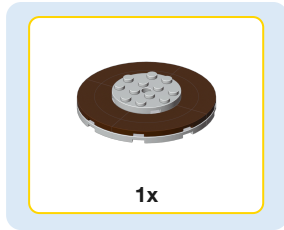


Hub with its drive teeth



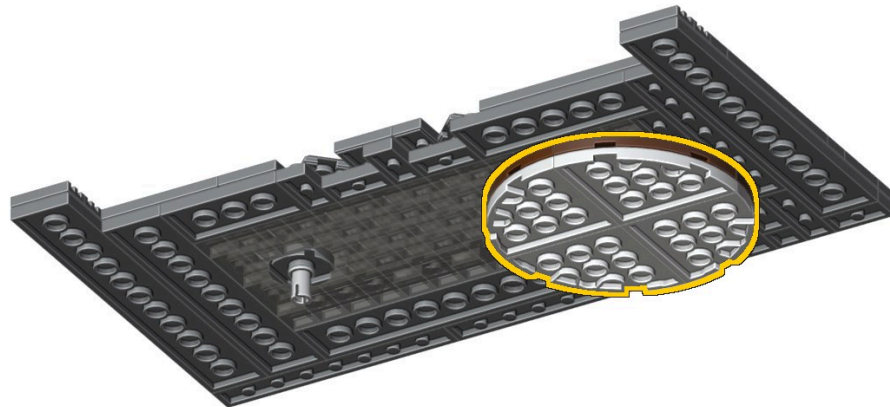
TOP HALF

75



seen from below

Slide the full supply reel onto its pin



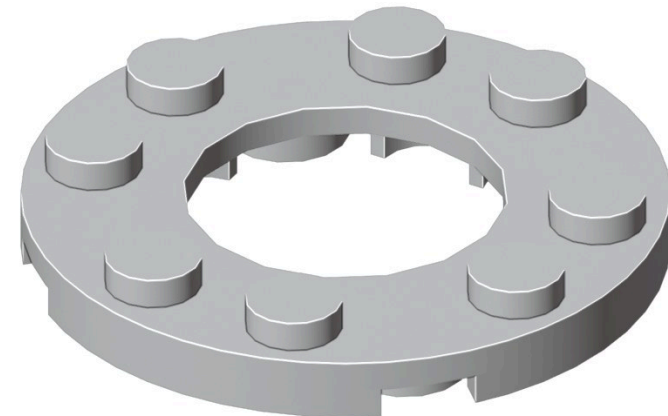
TAKE-UP REEL (NEARLY EMPTY)

76

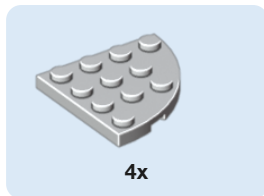


Take-up reel (nearly empty)

Hub core: the pin runs through its opening

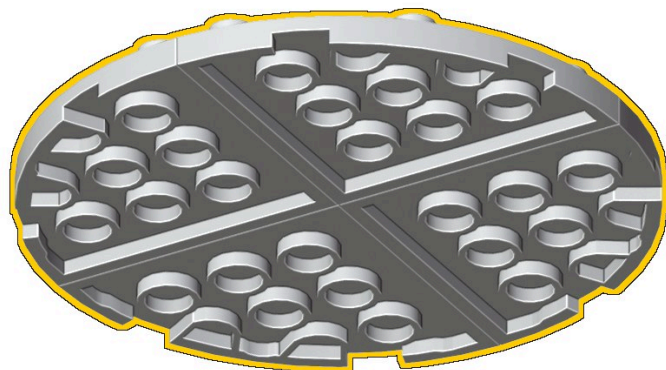


77

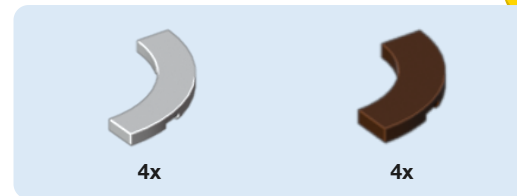


Lower flange from four quarter circles

seen from below

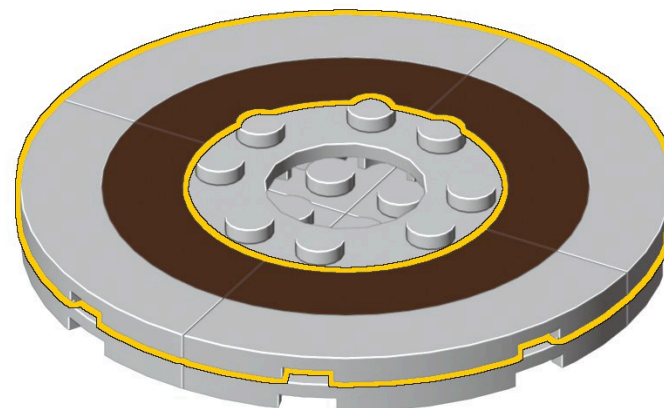


78



Just a few turns of tape left, so the flange shows

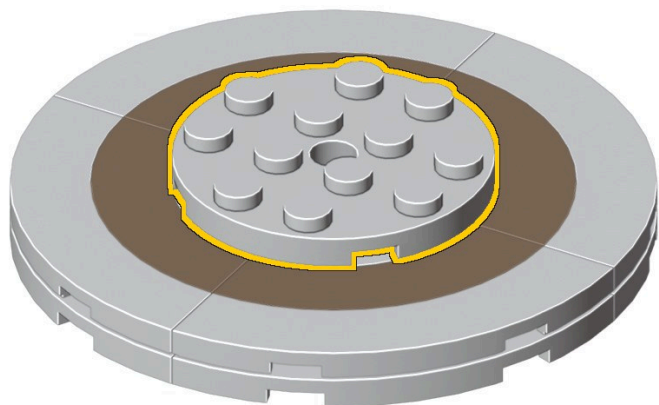
Take-up reel (nearly empty)



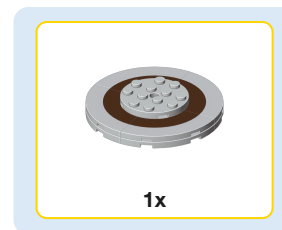
79



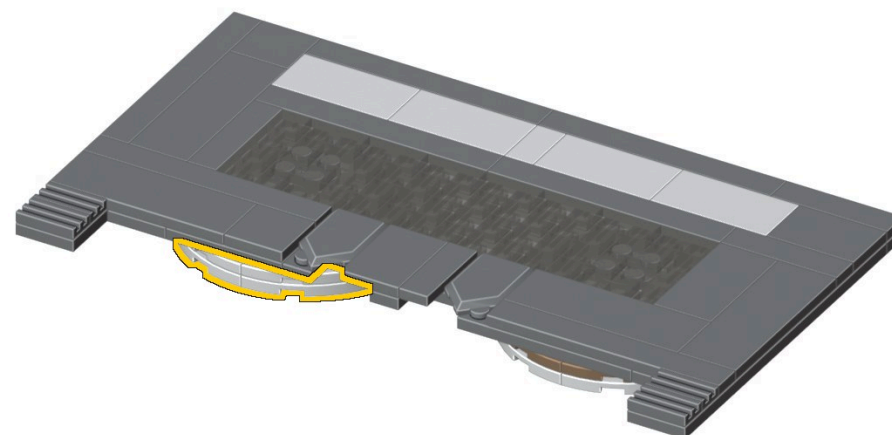
Hub with its drive teeth



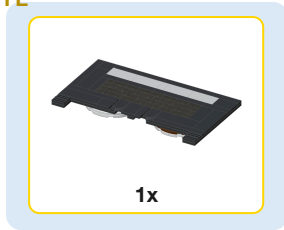
TOP HALF
80



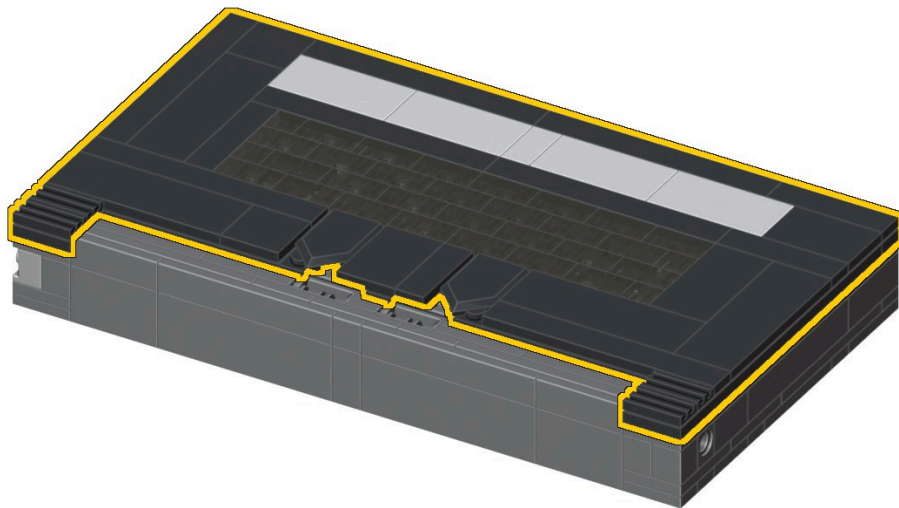
Slide the nearly empty take-up reel onto its pin



Top half



Lower the top half, reels and all, onto the bottom half

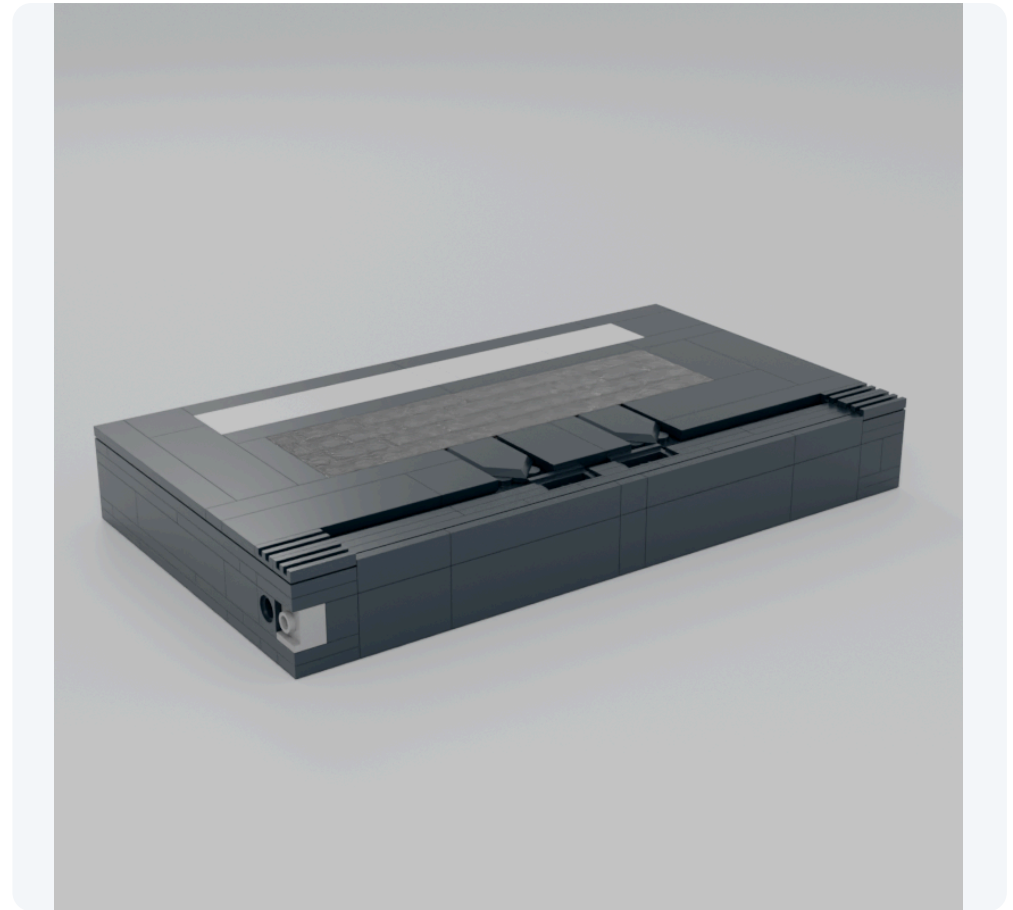


It works!



Dust door

The door swings up 90 degrees on two click hinges, over the tape run across the mouth.

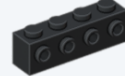
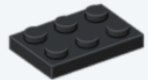
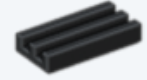


Spinning reels

Each reel turns on its own pin. Push the white flange through the round drive hole underneath and watch the tape pack turn in the window.

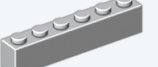
Parts inventory

348 pieces in 66 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 6x 300526 Black BrickLink 3005	 4x 300426 Black BrickLink 3004	 2x 362226 Black BrickLink 3622	 2x 300926 Black BrickLink 3009	 1x 4558954 Black BrickLink 87087	 4x 4162443 Black BrickLink 30414	 2x 393726 Black BrickLink 3937	 2x 4124096 Black BrickLink 6134
 16x 302426 Black BrickLink 3024	 12x 302326 Black BrickLink 3023	 11x 362326 Black BrickLink 3623	 17x 371026 Black BrickLink 3710	 8x 366626 Black BrickLink 3666	 15x 346026 Black BrickLink 3460	 1x 383226 Black BrickLink 3832	 2x 244526 Black BrickLink 2445
 1x 6001494 Black BrickLink 91988	 6x 302226 Black BrickLink 3022	 6x 302126 Black BrickLink 3021	 2x 302026 Black BrickLink 3020	 5x 379526 Black BrickLink 3795	 2x 303426 Black BrickLink 3034	 2x 4243819 Black BrickLink 3031	 2x 6168646 Black BrickLink 85861
 2x 403226 Black BrickLink 4032a	 2x 6052126 Black BrickLink 99206	 4x 654126 Black BrickLink 6541	 2x 307026 Black BrickLink 3070b	 3x 306926 Black BrickLink 3069b	 3x 4558170 Black BrickLink 63864	 3x 243126 Black BrickLink 2431	 2x 663626 Black BrickLink 6636
 12x 416226 Black BrickLink 4162	 3x 306826 Black BrickLink 3068b	 5x 6162892 Black BrickLink 26603	 1x 4560182 Black BrickLink 87079	 11x 6318582 Black BrickLink 69729	 8x 6434275 Black BrickLink 79393	 8x 6284070 Black BrickLink 98138	 4x 241226 Black BrickLink 2412b

Parts inventory (continued)

348 pieces in 66 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 2x 6245251 Black BrickLink 22385	 1x 4210719 Dark Bluish Gray BrickLink 3024	 11x 4633691 Flat Silver BrickLink 6141	 1x 4211476 Light Bluish Gray BrickLink 4070	 2x 4274194 Light Bluish Gray BrickLink 4274	 3x 6100769 Reddish Brown BrickLink 63864	 2x 4211194 Reddish Brown BrickLink 2431	 8x 6547793 Reddish Brown BrickLink 79393
 4x 6458167 Reddish Brown BrickLink 27507	 12x 6513960 Trans-Brown BrickLink 3024	 35x 6514006 Trans-Brown BrickLink 3023	 4x 6514259 Trans-Brown BrickLink 3070b	 28x 6514222 Trans-Brown BrickLink 3069b	 1x 6548257 White BrickLink 3004	 1x 6550143 White BrickLink 3622	 2x 6552097 White BrickLink 3009
 2x 6619283 White BrickLink 3062b	 1x 6548262 White BrickLink 3024	 2x 6552108 White BrickLink 3710	 1x 6550141 White BrickLink 3460	 2x 6615544 White BrickLink 11833	 2x 6564904 White BrickLink 60474	 8x 4178429 White BrickLink 30565	 1x 6548264 White BrickLink 3069b
 4x 6548271 White BrickLink 87079	 4x 6550142 White BrickLink 27507						

About this model

Two black halves close around two reels: a full supply reel and a nearly empty take-up reel, each spinning on its own pin behind a smoked window. The dust door clicks onto two hinges and swings up like the real flap. Look underneath for the drive holes, the screws, the write-protect tab and the open mouth where the head drum goes in.

Checked on a computer

PASS	real elements	66 part/colour combinations: 0 not real, 0 rare
PASS	connections	899 connections (882 stud, 13 pin, 2 hinge, 2 axle); 1 separate piece(s)
PASS	collisions	0 overlapping pair(s) among 348 parts
PASS	buildability	259 insertions tried across 6 submodel(s); 0 problem(s)
PASS	stability	about 288 g; centre of mass inside the base (100% margin); tips at 75.5 deg
PASS	mechanism	24 poses swept for 3 moving group(s); 0 gear pair(s); 0 problem(s)
PASS	electrics	no electrics in this model
PASS	technique	0 note(s)

Not yet built with real bricks

These checks catch most mistakes a designer can make, but not all: clutch power, part tolerances, how stiff a long axle is and how the model survives handling can only be judged with real bricks. If you build it, we would love to hear how it went.

Files

3D model (LDraw, opens in BrickLink Studio, LeoCAD and LDCad), parts lists for BrickLink and LEGO Pick a Brick, and an interactive 3D viewer:

lego.superfun.games/m/vhs_tape

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